

2020 Level I Mock Exam (A) AM

The 2020 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

- 1 Claire Jones, CFA, is an analyst following natural gas companies in the United States. At an industry energy conference, the CFO of Alpine Energy states they are interested in making strategic acquisitions. At a separate event, Alpine's head of exploration commented he is bullish on natural gas production prospects within Northeastern Pennsylvania. Jones is aware that Alpine currently has very little exposure to this region. She also knows another company in her universe, Pure Energy, Inc., is based in Northeastern Pennsylvania and controls significant assets in the area. Pure Energy is highly leveraged, and Jones believes they will need to raise additional capital or partner with another firm to move to the production phase with their assets. Jones attempts to contact Alpine's CEO with an unrelated question and is told he is unavailable because he is on a business trip to Northeastern Pennsylvania. Jones updates her research on Pure Energy and then recommends the stock to Lisa Wong, CFA, a portfolio manager who purchases significant positions in client accounts. The following week, Pure Energy announces that they have entered into an agreement to be purchased by Alpine for a significant premium. Has either Jones or Wong *most likely* violated Standards with regards to the integrity of capital markets?
- A No.
 - B Yes, Jones' recommendation is based on insider information.
 - C Yes, both Jones and Wong have acted on insider information.

A is correct because Jones has used the mosaic theory to combine non-material, non-public information with material public information.

B is incorrect because if taken in isolation, the information Jones received on the location of a business trip would be considered non-material.

C is incorrect because if taken in isolation, the information Jones received on the location of a business trip would be considered non-material.

Guidance for Standards I–VII

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- 2 Ian O’Sullivan, CFA, is the owner and sole employee of two companies, a public relations firm and a financial research firm. The public relations firm entered into a contract with Mallory Enterprises to provide public relations services, with O’Sullivan receiving 40,000 shares of Mallory stock in payment for his services. Over the next 10 days, the public relations firm issued several press releases that discussed Mallory’s excellent growth prospects. O’Sullivan, through his financial research firm, also published a research report recommending Mallory stock as a “buy.” According to the CFA Institute Standards of Professional Conduct, O’Sullivan is *most likely* required to disclose his ownership of Mallory stock in the:
- A press releases only.
 - B research report only.
 - C both the press release and the research report.

C is correct because members should disclose all matters that reasonably could be expected to impair the member’s objectivity [Standard I(B), Standard VI(A)].

A is incorrect because both the press release and the research report should disclose any potential conflict of interest.

B is incorrect because both the press release and the research report should disclose any potential conflict of interest.

Guidance for Standards I–VII

- 3 In order to provide investors with a more comprehensive view of a firm’s performance, the current GIPS standards includes new provisions related to:
- A various measures of risk.
 - B all aspects of performance measurement.
 - C the unique characteristics of each asset class.

A is correct. Historically, the GIPS standards focused primarily on returns. In the spirit of fair representation and full disclosure, and in order to provide investors with a more comprehensive view of a firm’s performance, the current GIPS standards includes new provisions related to risk.

B is incorrect, the GIPS standards do not address all aspects of performance measurement.

C is incorrect, the GIPS standards do not cover the unique characteristics of each asset class.

The GIPS Standards

- 4 When making performance presentations to prospective clients, a GIPS compliant firm should *least likely* do which of the following?
- A Selectively report its top fund performance
 - B Include the fund performance of former clients
 - C Report performance history for all market cycles under review



A is correct. When a firm complies with GIPS standards it cannot selectively choose its top fund performances while excluding weaker performing funds. It must include all fee-paying discretionary funds managed to a similar investment mandate, objective, or strategy.

B is incorrect because GIPS standards require the performance of former clients to be included to avoid survivorship bias.

C is incorrect because GIPS standards require the fund performance to reflect results across all market cycles for the periods under review. Fund managers are not allowed to select a time period during which the mandate produced superior results while leaving out other time period that underperformed (varying time periods).

Introduction to the Global Investment Performance Standards (GIPS)

- 5 For a retail client's portfolio to be included in a GIPS compliant firm's composite, it will *most likely* be in a composite:
- A composed of discretionary funds.
 - B restricted to retail funds.
 - C with both fee-paying and non-fee-paying funds.



A is correct. A composite must include all actual, fee-paying, discretionary portfolios managed in accordance with the same investment mandate, objective, or strategy.

B is incorrect because institutional funds may be included in a composite with retail funds if it has a similar investment mandate, objective, or strategy.

C is incorrect because non-fee-paying funds are not required to be in a GIPS compliant composite.

Introduction to the Global Investment Performance Standards (GIPS)

- 6 In countries where new local laws relating to calculation and presentation of investment performance conflict with GIPS standards, firms who have claimed GIPS compliance should *most likely*:
- A stop claiming GIPS compliance.
 - B follow local laws, continue to claim GIPS compliance, and disclose conflicts.
 - C continue to claim GIPS compliance, disclosing non-compliance with new laws.



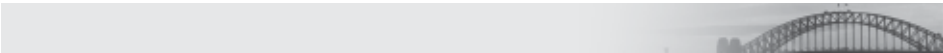
B is correct because where local laws and regulations regarding calculation and presentation conflict with GIPS standards, firms must abide by the local laws and regulations. They are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

A is incorrect because while firms must abide by local requirements firms are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

C is incorrect because while firms must abide by local requirements firms are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

Global Investment Performance Standards (GIPS)

- 7 Prudence Charmaine, a CFA charterholder, was recently accused in writing of cheating on a professional accounting exam. She denied cheating and successfully defended herself against the allegation. As part of her defense and as evidence of her character, Charmaine stated that she is a CFA charterholder and upholds the CFA Institute Code of Ethics and Standards of Professional Conduct. On her next annual Professional Conduct Statement, Charmaine does not report this allegation to CFA Institute. Did Charmaine *most likely* violate the CFA Institute Code of Ethics or Standards of Professional Conduct?
- A No
 - B Yes, she improperly used the CFA Institute Code and Standards to defend herself.
 - C Yes, she did not report the allegation on her annual Professional Conduct Statement.



C is correct because Charmaine should have reported the cheating allegation when making her annual Professional Conduct Statement. Even though she successfully defended herself against the charges and the charges were dropped, she has a responsibility to report the written complaint involving her integrity. The Code of Ethics requires CFA charterholders to practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

A is incorrect because Charmaine should have reported the cheating charges and the subsequent successful defense when making her annual Professional Conduct Statement. Even though she successfully defended herself against the charges, she has a responsibility to report the written complaint involving her integrity. The Code of Ethics requires CFA charterholders to practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

B is incorrect because it is not apparent that Charmaine violated Standard VII(B)—Reference to CFA Institute, the CFA Designation, and the CFA Program. Charmaine was correct in stating she is required to abide by the CFA Code and Standards.

CFA Institute Code of Ethics and Standards of Professional Conduct

- 8 Gardner Knight, CFA, is a product development specialist at an investment bank. Knight is responsible for creating and marketing collateralized debt obligations (CDOs) consisting of residential mortgage bonds. In the marketing brochure for his most recent CDO, Knight provided a list of the mortgage bonds that the CDO was created from. The brochure also states “an independent third party, the collateral manager, had sole authority over the selection of all mortgage bonds used as collateral in the CDO.” However, Knight met with the collateral manager and helped her select the bonds for the CDO. Knight is least likely to be in violation of which of the following CFA Institute Standards of Professional Conduct?
- A Suitability
 - B Conflicts of Interest

C Client Communication

A is correct because there is no indication that the investment is unsuitable for investors and in violation of Standard III(C)–Suitability.

B is incorrect because the conflict of interest represented when the bank selects the bonds making up the CDO, instead of the collateral manager, should be disclosed as required by Standard VI(A)–Disclosure of Conflicts.

C is incorrect because the basic format and general principles of the investment processes used to analyze investments, select securities, and construct portfolios should be disclosed as required by Standard V(B)–Communication with Clients and Prospective Clients, and any changes that might materially affect those processes must be promptly disclosed.

Guidance for Standards I–VII

- 9 Jan Loots, CFA, quit his job as a portfolio manager at an investment firm with whom he had a non-solicitation agreement he signed several years ago. Loots received permission to take his investment performance history with him and also took a copy of the firm's software-trading platform. Subsequently, Loots sent out messages on social media sites announcing he was looking for clients for his new investment management firm. Access to Loots' social media sites is restricted to friends, family, and former clients. Loots least likely violated the CFA Institute Standards of Professional Conduct concerning his:
- A trading software.
 - B non-solicitation agreement.
 - C investment performance history.

C is correct because the portfolio manager received permission to use his investment performance history from his prior employer. The member violated his non-solicitation agreement by indicating his availability to new clients on several social media sites accessible by clients of his former employer. This is a violation of Standard IV(A)–Loyalty because he did not act for the benefit of his former employer. In this case, the member may cause harm to his former employer if his weekend messages result in clients moving to his new business from his former employer. The member also violated this standard by taking his employer's property, trading software.

A is incorrect because the portfolio manager took property of his former employer, proprietary trading software, and violated Standard IV(A)–Loyalty. Although the manager created the software himself, it was during a period of time when the large money manager employed him and the software is not his property to take.

B is incorrect because the member violated Standard IV(A)–Loyalty as he did not act for the benefit of his former employer. In this case, the member may cause harm to his former employer if his weekend messages result in clients moving to his new business.

Guidance for Standards I–VII

- 10 Sherry Buckner, CFA, manages equity accounts for government entities whose portfolios are classified as being conservative and risk averse. Since the objective of her clients is to maximize returns with the lowest possible risk, Buckner considers adding to their holdings a new, thinly traded, leveraged derivative

product that she believes has the potential for high returns. To make her investment decision, Buckner relies upon comprehensive research from an investment bank with a solid reputation for top quality research. After her review of that research, Buckner positions her accounts so each has a 10% allocation to the derivative product. Did Buckner *most likely* violate any CFA Institute Standards of Professional Conduct by purchasing the derivative product for her clients?

- A No.
- B Yes, related to Suitability.
- C Yes, related to Loyalty, Prudence, and Care.

B is correct as Buckner is in violation of Standard III(C) since she did not consider issues such as the limited liquidity or any potential leverage of this new product when she invested a substantial percentage of her clients' portfolios in these instruments.

A is incorrect because Buckner violated the suitability Standard.

C is incorrect because Buckner relied upon comprehensive research from the investment bank.

Guidance for Standards I–VII

- 11 Wouter Duyck, CFA, is the sole proprietor of an investment advisory firm serving several hundred middle class retail clients. Duyck claims to be different from his competitors because he conducts research himself. He discloses that to simplify the management of all these accounts he has created a recommended list of stocks, from which he selects investments for all of his clients based on their suitability. Duyck's recommended list of stocks is obtained from his primary broker, who has completed due diligence on each stock. Duyck's recommended list *least likely* violates which of the following CFA Institute Standards of Professional Conduct?

- A Fair Dealing.
- B Misrepresentation.
- C Diligence and Reasonable Basis.

A is correct because Standard III(B)–Fair Dealing concerns the fair treatment of clients when making investment recommendations or taking investment action, but there is no indication that the advisor has discriminated against any clients with regard to his recommendations as he invests all clients in the same universe of stocks. The advisor has violated Standard I(C)–Misrepresentation with his research, which is not independently created and instead relies upon information provided by his broker. This is contrary to the advisor telling clients he does his own independent investment research. In addition, the advisor has violated Standard V(A)–Diligence and Reasonable Basis, as he has not made reasonable and diligent efforts to determine if the third party's research is sound.

B is incorrect, as the advisor has violated Standard I(C)–Misrepresentation with his research, which is not independently created and instead relies upon information provided by his broker.

C is incorrect, as the advisor has violated Standard V(A)–Diligence and Reasonable Basis as he does not have a reasonable basis for making his investment recommendations and relies solely on his broker’s research to create his list of stock investments. This is directly contrary to telling clients that he does his own independent investment research.

Guidance for Standards I–VII

- 12 Which of the following activities if undertaken by CFA Institute members and/or candidates would *most likely* violate the Code and Standards?
- A An analyst discloses confidential, sensitive information about a client account as part of an investigation by the CFA Institute Professional Conduct Program.
 - B A senior trader does not have safeguards in place to determine whether a junior trader under their supervision is following the firm’s policies regarding best execution.
 - C An institutional portfolio manager takes a group of clients to an expensive restaurant to discuss portfolio returns over the recently completed quarter without prior written consent from his employer.



B is correct because Standard IV(C)–Responsibilities of Supervisors states that members and candidates must make reasonable efforts to prevent violation of applicable laws, rules, regulations, and the Code and Standards by anyone subject to their supervision or authority. Interviewing with a competitor during lunch or taking clients out to lunch do not necessarily violate any Standard unless specifically prohibited in company policies.

A is incorrect because Standard III(E)–Preservation of Confidentiality is not intended to prevent a member or candidate from cooperating with an investigation with the CFA Institute Professional Conduct Program (PCP). Members and candidates can consider the PCP an extension of themselves when requested to provide information about a client in support of a PCP investigation.

C is incorrect because this action does not create a conflict with their employer. While it may be advisable to get prior approval before this event, this action does not constitute a clear violation of the Code and Standards as long as it does not conflict with any of the company’s policies.

Code of Ethics and Standards of Professional Conduct

- 13 Ken Kawasaki, CFA, shares a building with a number of other professionals who are also involved in the investment management business. Kawasaki makes arrangements with several of these professionals, including accountants and lawyers, to refer clients to each other. An informal score is kept on the expectation the referrals will equal out over time, eliminating the need for any cash payments. Kawasaki never mentions this arrangement to clients or prospective clients. Does Kawasaki’s agreement with the other building occupants *most likely* violate any CFA Institute Standards of Professional Conduct?
- A No.
 - B Yes, related to referral fees.
 - C Yes, related to communication with clients.



B is correct because Standard VI(C) requires disclosure of any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services. Even without cash changing hands the arrangement provides for a quid pro quo referral of clients and should be disclosed.

A is incorrect because Kawasaki has violated the Standards.

C is incorrect because this Standard has not been violated as it relates to disclosure to clients of the general principles of the investment process used and not disclosure of any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services as Standard VI(C) requires.

Guidance for Standards I–VII

- 14** Noor Hussein, CFA, runs a financial advisory business, specializing in retirement planning and investments. One of her clients asks her to advise the firm's pension fund trustees on available investments in the market including Islamic products. On the day prior to the meeting, Hussein spends an hour familiarizing herself with Islamic investment products and getting updates on local market conditions. The next day, she recommends Islamic investment products to the trustees based on her research and her expertise in retirement planning and investments. The trustees subsequently incorporate Islamic products into their investment allocation. Did Hussein's basis for the recommendation *most likely* comply with the CFA Code of Ethics?

- A** Yes.
- B** No, with regard to Misconduct.
- C** No, with regard to Diligence and Reasonable Basis.



C is correct because Hussein did not likely act with competence and diligence as required by Standard V(A). One half day of preparation with regard to Islamic investment products would not likely be considered sufficient to give investment advice to pension plan trustees. Misconduct was not violated by Hussein stating she is an expert in retirement planning and investments because this is the area she specializes in.

A is incorrect because Hussein did not likely act with competence and diligence as required by Standard V(A). One half day of preparation with regard to Islamic investment products would not likely be considered sufficient to give investment advice to pension plan trustees.

B is incorrect because it is not likely she violated Standard I(D)–Misconduct, i.e., conduct involving dishonesty, fraud, and/or deceit by stating she is an expert in retirement planning and investments.

Guidance for Standards I–VII

- 15** Alan Quanta, CFA, provides credit rating analysis of high-yield bonds using external credit ratings as a foundation. At the end of the last quarter, Quanta's firm, North Investment Bank, held a large position in the bonds of Veyron Corporation, a real estate company with all of its land holdings in a country recently downgraded by several credit rating agencies. The downgrades made Veyron bonds extremely difficult to sell because the bond price has dropped every day since the downgrades. Quanta has been asked by his supervisor to contact the firm's institutional clients to convince them Veyron bonds are still

an attractive purchase, especially at these lower prices. Quanta does not consider the Veyron bonds a buy at this price level. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, the *most* appropriate action for Quanta is to:

- A obey his supervisor's request.
- B ignore his supervisor's request.
- C promote the bonds with appropriate disclosures.

B is correct because Quanta must refuse to promote Veyron bonds until they are an attractive purchase based on fundamental analysis and market pricing. If Quanta followed the request from his supervisor, he would be in violation of Standard I(B)–Independence and Objectivity, as he does not rate Veyron bonds as a buy. His opinion of the Veyron bonds must not be affected by internal pressure or compensation.

A is incorrect because Quanta should refuse to follow his supervisor's request and promote the bonds as his opinion of the Veyron bonds must not be affected by internal pressure or compensation.

C is incorrect because Quanta should refuse to promote the bonds as his opinion of the Veyron bonds must not be affected by internal pressure or compensation.

Guidance for Standards I–VII

- 16 “Additional Compensation Agreements” is *most likely* a sub-section of which CFA Institute Standard of Professional Conduct?

- A Duties to Employers
- B Duties to Clients
- C Professionalism

A is correct. Standard IV–Duties to Employers includes a sub-section entitled “Additional Compensation Agreements”.

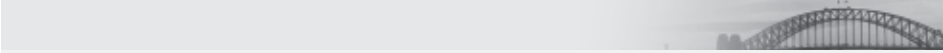
B is incorrect. Standard III–Duties to Clients does not include this sub-section.

C is incorrect. Standard I–Professionalism does not include this sub-section.

Code of Ethics and Standards of Professional Conduct

- 17 Joanne Bryce recently received notification she had successfully completed the Level III CFA exam. Having met all the necessary requirements, Bryce would soon be awarded her charter. Bryce's employer wanted to recognize her accomplishment and placed an ad in the local newspaper. Which of the following statements would *least likely* comply with the CFA Institute Standards of Professional Conduct?

- A The CFA charter is a key asset in the development of her investment career.
- B By becoming a charterholder, she has significantly improved her standing within the firm.
- C Ms. Bryce passed all three levels of the exam consecutively, placing her in an elite group.

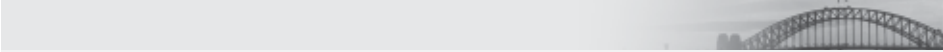


C is correct. The ad should not include the statement, “Ms. Bryce passed all three levels of the exam consecutively, placing her in an elite group.” CFA Standard VII(B): References to CFA Institute, the CFA Designation, and the CFA Programs states, “When referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, Members and Candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program.” CFA charterholders are those individuals who have earned the right to use the CFA designation granted by CFA Institute. These individuals have satisfied certain requirements, including completion of the CFA Program and the required years of acceptable work experience. It would not have been a violation to simply state that Ms. Bryce had consecutively passed all three levels of the exam; it would simply be a statement of fact. Adding that it places her in an elite group is an exaggeration of her accomplishment and could be viewed as misleading.

A and B are incorrect. The statements “The CFA charter is a key asset in the development of her investment career” and “By becoming a charterholder, she has significantly improved her standing within the firm” can be included in the ad. The charter can be considered a key asset, and that it has significantly improved her standing within the firm simply indicates that her firm holds the charter in high regard. Neither statement misrepresents nor exaggerates the meaning or implication of holding the CFA designation.

Guidance for Standards I-VII

- 18 Trust is *most likely* the foundation of the financial industry because:
- A financial products and services are often tangible and verifiable.
 - B investors rely on the specialized knowledge of investment professionals.
 - C global financial markets and their participants are highly regulated by effective laws.



B is correct. Trust is the foundation of the financial industry because investors rely on the specialized knowledge of investment professionals, which allows them to have more power than their clients with regard to making investment decisions. As a result, investors must trust their advisers to use their specialized knowledge to benefit their clients first and foremost.

A is incorrect because many financial products and services are intangible, so investors heavily rely on investment professionals to act in their best interests and to inform them honestly about their financial positions.

C is incorrect because most financial markets worldwide and their participants are highly regulated through effective laws. However, despite this regulation, unethical behavior can still occur because legal actions can still be unethical. Unethical behavior has a negative impact on investors. Therefore, trust depends on financial professionals acting in an ethical manner.

Ethics and Trust in the Investment Profession

- 19 Over the past four years, a portfolio experienced returns of –8%, 4%, 17%, and –12%. The geometric mean return of the portfolio over the four-year period is *closest* to:
- A 0.25%.
 - B –0.37%.

C 0.99%.

B is correct. Add one to each of the given returns, then multiply them together and take the fourth root of the resulting product. $0.92 \times 1.04 \times 1.17 \times 0.88 = 0.985121$; 0.985121 raised to the 0.25 power is 0.996259. Subtracting one and multiplying by 100 gives the correct geometric mean return: $[(0.92 \times 1.04 \times 1.17 \times 0.88)^{0.25} - 1] \times 100 = -0.37\%$.

A is incorrect because it is the arithmetic mean of the four numbers.

C is incorrect because it is the solution to: $(0.92 \times 1.04 \times 1.17 \times 0.88) = 0.99$ (rounded).

Organizing, Visualizing, and Describing Data

- 20 The following 10 observations are a sample drawn from an approximately normal population:

Observation	1	2	3	4	5	6	7	8	9	10
Value	-3	-11	3	-18	18	20	-6	9	2	-16

The sample standard deviation is *closest* to:

- A 13.18.
B 11.92.
C 12.50.

A is correct. The sample mean is:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$= (-3 - 11 + 3 - 18 + 18 + 20 - 6 + 9 + 2 - 16)/10$$

$$= -2.00/10 = -0.20$$

The sample variance is:

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}$$

The sample standard deviation is the (positive) square root of the sample variance.

Value	Difference vs. Mean [Value - (-0.20)]	Difference Squared
-3	-2.8	7.84
-11	-10.8	116.64
3	3.2	10.24
-18	-17.8	316.84
18	18.2	331.24
20	20.2	408.04
-6	-5.8	33.64

(continued)

Value	Difference vs. Mean [Value – (–0.20)]	Difference Squared
9	9.2	84.64
2	2.2	4.84
–16	–15.8	249.64
Sum of squared differences		1563.6
Divided by $n - 1$		173.7333333
Square root		13.18079411

B is incorrect and is calculated by dividing by 11 rather than 9

C is incorrect and is calculated by dividing by 10 rather than 9.

Organizing, Visualizing, and Describing Data

Cumulative Probabilities for a Standard Normal Distribution

$P(Z \leq x) = N(x)$ for $x \geq 0$ or $P(Z \leq z) = N(z)$ for $z \geq 0$

x or z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.10	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.20	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.30	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
...										
1.90	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.00	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.10	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
...										
2.50	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.60	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.70	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974

- 21 A variable is normally distributed with a mean of 5.00 and a variance of 4.00. Using the excerpt above from the cumulative distribution function for the standard normal random variable table, the probability of observing a value of –0.40 or less is *closest* to:
- A 2.44%.
- B 8.85%.
- C 0.35%.

C is correct. First the outcome of interest, –0.40, is standardized for the given normal distribution:

$$Z = (X - \mu)/\sigma = (-0.40 - 5.00)/2 = -2.70$$

Then use the table to find the probability of a Z value being 2.70 standard deviations *below* the mean (i.e., when $z \leq 0$). The value is $1 - P(Z \leq +2.70)$. In this problem, the solution is: $1 - 0.9965 = 0.0035 = 0.35\%$.

A is incorrect; it inverts mean and variance: $(-0.40 - 4.00)/\sqrt{5} = -1.97$ (z-value). The probability is then calculated as: $1 - 0.9756 = 2.44\%$.

B is incorrect; it divides -5.4 by the variance, 4, and uses 1.35 as the z-value. The probability is then calculated as: $1 - 0.9115 = 8.85\%$.

Common Probability Distributions

- 22 A small-cap growth fund's monthly returns for the past 36 months have been consistently outperforming its benchmark. An analyst is determining whether the standard deviation of monthly returns is greater than 6%. Which of the following *best* describes the hypothesis to be tested?

- A $H_0: \sigma^2 \leq 0.36\%$
 B $H_a: \sigma^2 > 6\%$
 C $H_0: \sigma^2 \geq 0.36\%$

A is correct. This is a one-tailed hypothesis testing with a "greater than" alternative hypothesis. A squared standard deviation is being used to obtain a test of variance. The hypotheses are $H_0: \sigma^2 \leq 0.36\%$ versus $H_a: \sigma^2 > 0.36\%$.

B is incorrect as explained in choice A.

C is incorrect as explained in choice A

Hypothesis Testing

- 23 The following information is available for a portfolio:

Asset Class	Asset Allocation Weight (%)	Asset Class Return (%)	Correlation with Equities Class (%)
Equities	45	16	100
Mortgages	25	12	30
Cash and equivalents	30	2	10

The return on the portfolio is *closest* to:

- A 10.0%.
 B 8.2%.
 C 10.8%.

C is correct. The portfolio return is the weighted mean return and is calculated as:

$$\begin{aligned}\bar{X}_w &= \sum_{i=1}^n w_i X_i \\ &= 0.45 \times 16 + 0.25 \times 12 + 0.30 \times 2 \\ &= \mathbf{10.80}.\end{aligned}$$

B is incorrect. It weights each component by the correlation with equities: $0.45 \times 16 \times 1.00 + 0.25 \times 12 \times 0.30 + 0.30 \times 2 \times 0.10 = 8.16$.

A is incorrect. It is the arithmetic average of the three returns without weights: $(16 + 12 + 2)/3 = 10$.

Organizing, Visualizing, and Describing Data

- 24** If the distribution of the population from which samples of size n are drawn is positively skewed and given that the sample size, n , is large, the sampling distribution of the sample means is *most likely* to have a:
- A** mean smaller than the mean of the entire population.
 - B** variance equal to that of the entire population.
 - C** distribution that is approximately normal.

C is correct. Given a population that has a finite variance and a large sample size, the central limit theorem establishes that the sampling distribution of sample means will be approximately normal, will have a mean equal to the population mean, and will have a variance equal to the population variance divided by the sample size.

A is incorrect. Given a population that has a finite variance and a large sample size, the central limit theorem establishes that the sampling distribution of sample means will have a mean *equal* to the population mean.

B is incorrect. Given a population that has a finite variance and a large sample size, the central limit theorem establishes that the sampling distribution of sample means will have a variance equal to the population variance *divided by the sample size*.

Sampling and Estimation

- 25** A fund manager would like to estimate the probability of a daily loss higher than 5% on the fund he manages. He decides to use a method that uses the relative frequency of occurrence based on historical data. The resulting probability is *best* described as a(n):
- A** subjective probability.
 - B** a priori probability.
 - C** empirical probability.

C is correct. An empirical probability is a probability estimated from data as a relative frequency of occurrence.

A is incorrect. A subjective probability is a probability drawing on personal or subjective judgment.

B is incorrect. An a priori probability is a probability obtained based on logical analysis.

Probability Concepts

Population	1	2
Sample size	$n_1 = 5$	$n_2 = 5$
Sample variance	$s_1^2 = 4$	$s_2^2 = 28$

- 26 The samples are drawn independently and both populations are assumed to be normally distributed

Using the above data, an analyst is trying to test the null hypothesis that the population variances are equal ($H_0: \sigma_1^2 = \sigma_2^2$) against the alternative hypothesis that the variances are not equal ($H_a: \sigma_1^2 \neq \sigma_2^2$) at the 5% level of significance. The table of the F -distribution is provided below.

Table of the F -Distribution

Panel A: Critical values for right-hand tail area equal to 0.05

		df1 (read across)				
		1	2	3	4	5
df2 (read down)	1	161	200	216	225	230
	2	18.5	19.0	19.2	19.2	19.3
	3	10.1	9.55	9.28	9.12	9.01
	4	7.71	6.94	6.59	6.39	6.26
	5	6.61	5.79	5.41	5.19	5.05

Panel B: Critical values for right-hand tail area equal to 0.025

		df1 (read across)				
		1	2	3	4	5
df2 (read down)	1	648	799	864	900	922
	2	38.51	39.00	39.17	39.25	39.30
	3	17.44	16.04	15.44	15.10	14.88
	4	12.22	10.65	9.98	9.60	9.36
	5	10.01	8.43	7.76	7.39	7.15

Which of the following statements is *most* appropriate? The critical value is:

- A 6.39 and reject the null.
- B 9.60 and do not reject the null.
- C 7.15 and do not reject the null.

B is correct. The test statistic makes use of the F -distribution and is the ratio of the variances, with the larger variance in the numerator. The test statistic is $F = s_2^2 / s_1^2 = 28/4 = 7$. The degrees of freedom are 4 by 4. As it is a two-tailed test, the correct critical value at $\alpha = 5\%$ is 9.60 (Panel B). As the test statistic is less than the critical value (i.e., $7 < 9.60$), the null hypothesis cannot be rejected.

A is incorrect. It uses 4 by 4 degrees of freedom and does not divide the given alpha by two as is appropriate for a two-tailed test.

C is incorrect. It uses the 5 by 5 critical value (i.e., 5 degrees of freedom) rather than correctly using the 4 by 4 value (i.e., 4 degrees of freedom).

Hypothesis Testing

- 27 A z-test regarding a mean is *most* appropriate when the variance is unknown and the:
- A sample is not normally distributed.
 - B central limit theorem is applicable.
 - C sample is small and normally distributed.

B is correct. If the population variance is unknown and the sample is large, we can use a z-test relying on the central limit theorem.

A is incorrect because a *t*-test is appropriate in the case of an unknown variance with moderate departures from normality.

C is incorrect because in the case of a hypothesis test of the population mean with an unknown variance, a test statistic is used when either 1) the sample is large, or 2) the sample is small but normally distributed, or approximately normally distributed. A *t*-test is typically used with small samples and an unknown population variance.

Hypothesis Testing

- 28 Which of the following tests should be used to evaluate the difference between the means of two normally distributed populations?
- A An approximate *t*-test if the population variances are unknown and assumed unequal and the samples are assumed to be independent.
 - B An approximate *t*-test that involves the calculation of a pooled estimator of the population variances which are assumed unequal.
 - C A paired comparison test if the two samples are independent and the population variances are unknown.

A is correct. An approximate *t*-test is used to test the differences between means of two populations when the unknown population variances cannot be assumed to be equal.

B is incorrect because a pooled estimator is used when the two population variances are assumed to be equal.

C is incorrect because a paired comparison test is used to test mean differences when the samples are assumed to be dependent.

Hypothesis Testing

- 29 A random number between zero and one is generated according to a continuous uniform distribution. What is the probability that the first number generated will have a value of exactly 0.30?
- A 70%
 - B 0%

C 30%

B is correct. The probability of generating a random number equal to any fixed point under a continuous uniform distribution is zero.

C is incorrect because this is the cumulative distribution function of the continuous uniform distribution. The probability $P(0 \leq x \leq 0.3)$ in a continuous uniform distribution with $a = 0$ and $b = 1$ is equal to:

$$F(0.3) = \frac{x - a}{b - a} = \frac{0.3 - 0}{1 - 0} = 0.3$$

A is incorrect because this is the probability of a random number x in this continuous uniform distribution taking a value greater than or equal to 0.3; $P(x \geq 0.3) = 1 - F(0.3) = 0.7$.

Common Probability Distributions

- 30** A company has reported annual revenue for the past five years as follows (£ millions):

2008	2009	2010	2011	2012
250,738	279,347	316,480	355,589	392,008

The company's compound annual growth rate in revenue is *closest* to:

- A** 11.8%.
B 14.1%.
C 9.4%.

A is correct. Compound growth rate can be calculated as

$$\left[\left(\frac{\text{Ending value}}{\text{Beginning value}} \right)^{1/\text{number of periods}} - 1 \right] = \left[\left(\frac{392,008}{250,738} \right)^{1/4} - 1 \right] = 11.8\%$$

B is incorrect because it determines the average growth rate: $[(392,008/250,738) - 1]/4 = 14.1\%$.

C is incorrect because it uses 5 periods instead of 4: $[(392,008/250,738)^{1/5} - 1] = 9.35\%$.

The Time Value of Money

- 31** The following information applies to a start-up company solely owned by an entrepreneur.

	Value
Total units produced	3,550
Average revenue	\$1,110
Average variable cost	\$750
Total fixed cost	\$300,000
Total investment	\$1,550,000

(continued)

	Value
Required rate of return	12.5%
Opportunity cost of owner's labor	\$125,000

The company's economic profit is *closest to*:

- A \$659,250.
- B \$784,250.
- C \$318,750.

A is correct.

Economic profit = Accounting profit – Total implicit opportunity costs
where

Accounting profit = Total revenue – Total variable costs – Total fixed costs

Total opportunity costs = opportunity cost of capital + opportunity cost of labor

Total revenue	$3,550 \times \$1,110$	\$3,940,500	# units $\times$ average revenue
Less Total variable costs	$3,550 \times \$750$	\$2,662,500	# units $\times$ average var cost
Less Total fixed costs		\$300,000	given
Accounting profit		\$978,000	
Opportunity cost of capital	$\$1,550,000 \times 0.125$	\$193,750	Investment $\times$ Required return
Opportunity cost of owner's labor		\$125,000	Given
Total opportunity costs		\$318,750	
Economic profit		\$659,250	

C is incorrect because it calculates the normal profit or the total implicit opportunity cost.

B is incorrect because it does not take into account the opportunity cost of labor. Total implicit opportunity cost = $\$1,550,000 \times 0.125 = \$193,750$. Economic profit = $\$978,000 - \$193,750 = \$784,250$.

The Firm and Market Structures

- 32 If the prices of substitute resources decrease, the demand for a given resource will *most likely*:
- A remain unchanged.
 - B decrease.
 - C increase.

B is correct. A decrease in the price of a substitute resource would encourage producers to use the substitute resource thus reducing demand for the resource in question.

C is incorrect. A decrease in the price of a substitute good would induce consumers to use the substitute good, reducing demand for the good in question.

A is incorrect. A decrease in the price of a substitute good would induce consumers to use the substitute good, reducing demand for the good in question.

Topics in Demand and Supply Analysis

33 The following information applies to a hypothetical economy:

Total population	1,100
Working age population	975
Labor force	750
Underemployed	120
Unemployed	95
Discouraged workers	80
Frictionally unemployed	25
Voluntarily unemployed	40

The unemployment rate is *closest* to:

- A** 12.7%.
- B** 16.0%.
- C** 9.7%.

A is correct. Unemployment rate = $(\text{Unemployed} / \text{Labor force}) \times 100 = (95 / 750) \times 100 = 12.7\%$.

B is incorrect. It includes the frictionally unemployed (which is already part of the unemployed): $(95 + 25) / 750 \times 100 = 16.0\%$

C is incorrect. It compares the unemployed to the working age population: $95 / 975 \times 100 = 9.7\%$.

Understanding Business Cycles

34 Based on the elasticities approach, a country can implement an exchange rate policy to improve its trade balance *most* effectively if it imports and exports products:

- A** that are consumer necessities.
- B** with no good substitute.
- C** traded in competitive markets.

C is correct. In the elasticities approach, changes in exchange rate policy will be a more-effective mechanism for trade balance adjustment if a country imports and exports products that trade in competitive markets, with good substitutes, and luxury products rather than necessities.

A is incorrect. In the elasticities approach, changes in exchange rate policy will be a more-effective mechanism for trade balance adjustment if a country imports and exports products that trade in competitive markets, with good substitutes, and luxury products rather than necessities.

B is incorrect. In the elasticities approach, changes in exchange rate policy will be a more-effective mechanism for trade balance adjustment if a country imports and exports products that trade in competitive markets, with good substitutes, and luxury products rather than necessities.

Currency Exchange Rates

- 35** Assuming its trading partner does not retaliate, which of the following conditions *must* hold in order for a large country to increase its national welfare by imposing a tariff?
- A** The deadweight loss must be smaller than the benefit of its improving terms of trade.
 - B** It must auction the import licenses for a fee to offset the decline in the consumer surplus.
 - C** It must have a comparative advantage in the production of the imported good.



A is correct. The large country is able to cause the foreign exporter to reduce price in order to retain market share. In the large country, domestic producers gain from higher volume and the government gains from collecting the tariff. The sum of these two gains must exceed the deadweight loss to domestic consumers to achieve a national welfare gain. The change in terms of trade causes income redistribution from the foreign exporter to the domestic producer.

B is incorrect. An import license relates to a quota, not a tariff.

C is incorrect. If the large country had a comparative advantage, it would be exporting more than importing. This is not relevant to whether there is a net domestic gain from the tariff. The tariff hurts domestic consumers. Unless the gain from the tariff exceeds the loss to consumers, national welfare will decrease.

International Trade and Capital Flows

- 36** A central bank announcement of a program to raise rates to moderate inflation will *most likely* lead to:
- A** a weaker domestic currency.
 - B** revised interest rate expectations.
 - C** higher asset prices.



B is correct. Companies and individuals often make investment and purchasing decisions based on interest rate expectations extrapolated from recent events. As the economic actors perceive that a central bank will be implementing rate increases, they will anticipate rising rates and adjust their behavior accordingly. These revised higher interest rate expectations will typically lead to a stronger domestic currency and declines in consumption, asset prices, and borrowing. These outcomes are consistent with a policy action to moderate inflation.

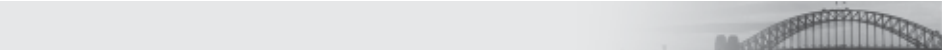
A is incorrect because raising central bank rates to moderate inflation typically leads to a stronger (rather than a weaker) domestic currency. The strengthening domestic currency makes domestic exports more expensive to foreign buyers thereby dampening demand, an outcome that is consistent with the objective of moderating inflation.

C is incorrect because raising central bank rates to moderate inflation typically leads to declining asset prices as the discount rate rises. Declining asset prices lead market participants to a view that higher interest rates result in slower economic growth, reduced profits, and reduced borrowing to finance asset purchases. Reduced consumer demand from the wealth effect of declining asset prices also tempers consumption and inflationary pressures.

Monetary and Fiscal Policy

37 The structural deficit is equal to the budget deficit:

- A** adjusted for inflation.
- B** that would exist at full employment.
- C** excluding the impact of automatic stabilizers.



B is correct. The structural deficit is the deficit that would exist if the economy was at full employment (or full potential output). Economists often consider the structural deficit as an indicator of the fiscal policy stance.

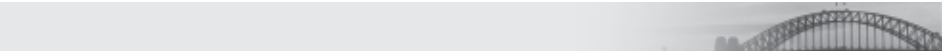
A is incorrect because the structural deficit makes no adjustment for inflation.

C is incorrect because the structural deficit includes (rather than excludes) the impact of automatic stabilizers on the budget assuming full employment.

Monetary and Fiscal Policy

38 Both monetary and fiscal policies can *most likely* be used by a government to:

- A** redistribute income and wealth.
- B** affect the level of interest rates.
- C** influence the level of economic activity.



C is correct. Both monetary and fiscal policies are used to regulate economic activity over time.

A is incorrect. Only fiscal policy can be used to redistribute income and wealth.

B is incorrect. Only monetary policy can be used to affect the level of interest rates.

Monetary and Fiscal Policy

39 Which statement about the Ricardian model of trade is *most* accurate? The model:

- A** allows for a redistribution of income through trade.
- B** incorporates the benefit of technology in labor productivity.
- C** predicts that a country with abundant labor resources relative to capital will likely export labor-intensive goods and import capital-intensive goods.



B is correct. In the Ricardian model of trade, labor is the only input and the benefit of technology is reflected in labor productivity.

A is incorrect. Income redistribution cannot happen in a Ricardian model because capital is not an input in the Ricardian model. Income redistribution can happen in the Heckscher–Ohlin model of trade because the model incorporates capital and labor.

C is incorrect. The Ricardian model does not have capital as an input and consequently cannot predict trade on the basis of labor-intensive goods and capital-intensive goods. Trade predictions based on capital and labor can be determined using the Heckscher–Ohlin model.

International Trade and Capital Flows

- 40 An exchange rate between two currencies has decreased to 1.3500. If the base currency has depreciated by 7% against the price currency, the initial exchange rate between the two currencies was *closest* to:

- A 1.2617.
- B 1.4445.
- C 1.4516.



C is correct. To calculate the initial exchange rate, the current exchange rate is divided by 1 minus the percentage depreciation in the exchange rate of the base currency: $1.3500 \div (1 - 0.07) = 1.3500 \div 0.93 = 1.4516$.

A is incorrect. The calculation incorrectly divides the current exchange rate by 1 plus the percentage depreciation in the exchange rate: $1.3500 \div 1.07 = 1.2617$.

B is incorrect. The calculation incorrectly multiplies the current exchange rate by 1 plus the percentage depreciation in the exchange rate: $1.3500 \times 1.07 = 1.4445$.

Currency Exchange Rates

- 41 Which of the following statements about a downward-sloping long-run average cost (LRAC) curve is *most* accurate? A downward-sloping LRAC curve is representative of a firm experiencing:

- A economies of scale.
- B diseconomies of scale.
- C decreasing levels of investment.



A is correct. When the LRAC curve is downward sloping, it means the firm is producing units at lower average costs per unit as production levels rise. This situation represents economies of scale in production.

B is incorrect. An upward-sloping LRAC curve would be representative of diseconomies of scale.

C is incorrect. Decreasing levels of investment alone would not result in a downward-sloping LRAC curve; it would also depend on the volume of production. If the level of production drops proportionately more than the level of investment, the LRAC curve could be upward sloping.

Topics in Demand and Supply Analysis

42 Selected information is taken from the GDP data of a hypothetical country:

Account Name	Amount (\$ trillions)
GDP	18.0
Wages, salaries, and other labor income	9.0
Taxes	3.5
Capital consumption allowance	2.3
Undistributed corporate profits	1.0
Transfer payments	1.9

Personal income (in \$ trillions) for this country is *closest* to:

- A 10.9.
- B 15.4.
- C 17.7.

C is correct. Personal income = National income – Taxes – Undistributed corporate profits + Transfer payments, where

$$\text{National income} = \text{GDP} + \text{Capital consumption allowance} = 18 + 2.3 = 20.3$$

$$\text{Personal income} = 20.3 - 3.5 - 1.0 + 1.9 = 17.7$$

A is incorrect. It is only cash flows to households through wages, salaries, and other labor income plus transfer payments = $9.0 + 1.9 = 10.9$.

B is incorrect. It ignores capital consumption allowance in determining national income. National income is incorrectly calculated as

$$\text{National income} = \text{GDP} + \text{Capital consumption allowance} = 18 + 0 = 18, \text{ with}$$

$$\begin{aligned} \text{Personal income} &= \text{National income} - \text{Taxes} - \text{Undistributed corporate} \\ &\quad \text{profits} + \text{Transfer payments} \\ &= 18.0 - 3.5 - 1.0 + 1.9 = 15.4. \end{aligned}$$

Aggregate Output, Prices, and Economic Growth,"

43 Under International Financial Reporting Standards (IFRS), which of the following is *most likely* one of the general features underlying the preparation of financial statements?

- A Understandability
- B Timeliness
- C Consistency

C is correct. Consistency is one of the general features underlying the preparation of financial statements based on IFRS.

A is incorrect. Understandability is one of the qualitative characteristics of financial statements under IFRS framework for the preparation and presentation of financial statements. It is not a general feature.

B is incorrect. Timeliness is one of the qualitative characteristics of financial statements under IFRS framework for the preparation and presentation of financial statements. It is not a general feature.

Financial Reporting Standards

- 44 A company acquires a license for \$6,500 with the right to use the license for four years. Management expects to derive benefits from the license for three years and uses the straight-line amortization method. Accumulated amortization at the end of Year 2 is *closest to*:
- A \$4,333.
 - B \$3,250.
 - C \$2,167.

A is correct. Accumulated amortization for the intangible asset at the end of Year 2 is closest to \$4,333. At the end of the second year, amortization taken = 2 years $\times$ ($6,500/3$) = \$4,333.

B is incorrect. It mistakenly uses the legal life of four years instead of three years. This gives an amortization expense of $6,500/4 = \$1,625$ per year. Using a four-year useful life, accumulated amortization at the end of Year 2 is calculated as 2 years $\times$ \$1,625 = \$3,250.

C is incorrect. It calculates amortization expense in Year 2, $6,500/3 = \$2,167$ but forgets to add the amortization expense from Year 1 to arrive at accumulated amortization.

Long-Lived Assets

- 45 Under IFRS, the costs incurred in the issuance of bonds are *most likely*:
- A expensed when incurred.
 - B included in the measurement of the bond liability.
 - C deferred as an asset and amortized on a straight-line basis.

B is correct. Under IFRS, debt issuance costs are included in the measurement of the bond liability.

A is incorrect. Under both US GAAP and IFRS, they are not expensed.

C is incorrect. This is US GAAP.

Non-Current (Long-Term) Liabilities

- 46 In the current year, a company increased its deferred tax asset by \$500,000. During the year, the company *most likely*:

- A became entitled to a \$500,000 tax refund.
- B had permanent differences between accounting profit and taxable income.
- C reported a lower accounting profit than taxable income.

C is correct. Deferred tax assets represent taxes that have been paid (because of the higher taxable income) but have not yet been recognized on the income statement (because of the lower accounting profit).

A is incorrect. Deferred tax assets are simply the results of differences between accounting profit and taxable income. It is not an amount of a tax refund that would be an income tax receivable.

B is incorrect. Only temporary differences create deferred tax assets or liabilities

Income Taxes

- 47 A company acquires some new depreciable assets. It uses straight-line depreciation for all of its assets. Which of the following combinations of estimated residual values and useful lives is *most likely* to produce the highest net profit margin? Estimated residual values should be:
- A high with long average lives.
 - B low with long average lives.
 - C high with short average lives.

A is correct. A high residual value estimate reduces the depreciable base and thus depreciation expense. Long average lives reduce the annual depreciation expense for any given depreciable base. The combination of the two would result in the lowest depreciation expense, which would lead to the highest net income and profit margins.

B is incorrect. Low residual values increase depreciation, though long lives reduce it.

C is incorrect. Short lives increase depreciation expense, though high residual values reduce it.

Long-Lived Assets

- 48 Which of the following suggestions is *best* aligned with CFA Institute advocacy for financial reporting that reflects economic reality?
- A Cash flow statements prepared using the direct format for cash flow from operations (CFO)
 - B Conservatism in revenue recognition policies
 - C Detailed cost information for long-lived assets

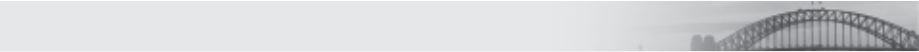
A is correct. The CFA Institute position paper "A Comprehensive Business Reporting Model: Financial Reporting for Investors" (2007) advocated for financial reporting that reflects economic reality. It specifically identified a preference for using the direct format for CFO in cash flow statements.

B is incorrect. The position paper advocated for neutrality in financial reporting.

C is incorrect. The position paper advocated for information regarding the current fair value of assets and liabilities, not cost information.

Financial Reporting Standards

- 49 The year-end balances in a company's last-in, first-out (LIFO) reserve are \$56.8 million as reported in the company's financial statements for both 2013 and 2014. For 2014, the measure that will most likely be the same regardless of whether the company uses the LIFO or the first-in, first-out (FIFO) inventory method is the:
- A gross profit margin.
 - B amount of working capital.
 - C inventory turnover.



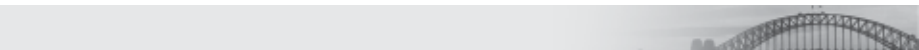
A is correct. The LIFO reserve did not change from 2013 to 2014. With no change in the LIFO reserve, cost of goods sold will be the same under both methods. Sales are always the same for both methods, so gross profit margin will be the same for 2014. The FIFO inventory will be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would differ under FIFO, both inventory turnover and the amount of working capital would also differ under FIFO.

B is incorrect. The FIFO inventory would be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would be different under FIFO, net working capital would be different under FIFO.

C is incorrect. The FIFO inventory would be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would be different under FIFO, the inventory turnover would be different under FIFO.

Inventories

- 50 Under IFRS, dividends received are *least likely* classified as which type of cash flow on the cash flow statement?
- A Financing
 - B Operating
 - C Investing



A is correct. Dividends received can be classified as either an operating or investing activity under IFRS but not as a financing activity.

B is incorrect because dividends received can be classified as an operating activity under IFRS.

C is incorrect because dividends received can be classified as an investing activity under IFRS.

Understanding Cash Flow Statements

- 51 The following information is available on a company:

Metric	
Fixed charge coverage ratio required by debt covenant	3.50
Forecasted interest expense (\$ thousands)	850
Forecasted interest payments (\$ thousands)	800
Forecasted lease payments (\$ thousands)	300
Tax rate	30%

The minimum net income (in thousands) that the company must generate to meet its debt covenant requirement is *closest* to:

- A \$2,700.
- B \$1,890.
- C \$2,100.

B is correct. The fixed charge coverage ratio is:

$$\frac{\text{Net income} + \text{Income tax expense} + \text{Interest expense} + \text{Lease payments}}{\text{Interest payments} + \text{Lease payments}}$$

	Calculation	\$ thousands
Denominator (\$ thousands): Interest payments + Lease payments	800 + 300	1,100
Minimum numerator: Net income + Income tax expense + Interest expense + Lease payments	Denominator × 3.50	3,850
Minus interest expense		-850
Minus lease payments		-300
Income before tax		2,700
Minus income tax expense	2,700 × 30%	-810
Minimum net income		1,890

A is incorrect. This answer incorrectly omits tax expense from the numerator. The answer provided is actually net income before tax as calculated above, not net income.

C is incorrect. This answer incorrectly assumes that the numerator in the ratio is EBIT.

(\$ thousands)

Minimum numerator (from above): Incorrectly assumed to be EBIT	Denominator × 3.50	3,850
Less: interest expense		-850
Incorrect income before tax		3,000
Less: income tax expense	\$3,000 × 30%	-900
Incorrect minimum net income		2,100

- A** Reducing the allowance for bad debt expense below the experienced loss rate.
- B** Deferring R&D expenses to a subsequent year.
- C** Choosing to depreciate new equipment over the shortest estimate of its useful life.

C is correct. Depreciating equipment over the shortest estimated period of its useful life is a conservative accounting choice that reduces earnings in the early years and increases them in the future, creating a positive trajectory.

A is incorrect. Reducing the bad debt allowance below the experienced loss rate is an aggressive choice that causes earnings to appear higher in the current year.

B is incorrect. Deferring R&D is an aggressive choice that causes earnings to appear higher in the current year.

Financial Reporting Quality

53 The following financial information is available at the end of the year.

Share Information

Security	Authorized	Issued and Outstanding	Other Features
Common stock	500,000	250,000	Currently pays a dividend of \$1 per share.
Preferred stock, Series A	50,000	12,000	Nonconvertible, cumulative; pays a dividend of \$4 per share.
Preferred stock, Series B	50,000	30,000	Convertible; pays a dividend of \$7.50 per share. Each share is convertible into 2.5 common shares.

Additional information:

Reported income for the year	\$1,000,000
------------------------------	-------------

The diluted EPS (earnings per share) is *closest* to:

- A** \$3.08.
- B** \$2.93.
- C** \$2.91.

C is correct. The convertible preferred shares are anti-dilutive, as shown in the following table. Therefore, the diluted EPS is the same as the basic EPS, \$2.91.

	Basic EPS	Diluted EPS (using if-converted method)	
Net income	\$1,000,000	\$1,000,000	
Preferred stock, Series A	(48,000)	(48,000)	12,000 shares × \$4/share
Preferred stock, Series B	(225,000)	0	30,000 shares × \$7.50/share
Earnings available to common shareholders	\$727,000	\$952,000	

Weighted Average Number of Common Shares (WACS)

Shares outstanding	250,000	250,000	
If converted	—	75,000	2.5 common/preferred × 30,000 preferred
WACS	250,000	325,000	
EPS = Earnings available to common shareholders/WACS	\$2.91	\$2.93*	

* Exceeds Basic EPS; Series B is anti-dilutive and is thus not included.

A is incorrect. It ignores the Series A dividends and incorrectly includes the anti-dilutive effect of converting the Series B shares.

From the table:

Earnings to Common = 1,000,000 (ignores Series A, converts)

Common shares outstanding = 325,000, as per table with conversion of convertible

EPS = \$1,000,000/325,000 shares = \$3.08/share

B is incorrect. It includes the conversion of the convertible securities, which are anti-dilutive.

Understanding Income Statements

54 A company's balance sheet at the end of the year shows the following:

Current Assets

Cash and cash equivalents	\$2,950
Marketable securities	730
Notes and accounts receivable, trade	5,740
Less allowance for doubtful accounts and sales returns	(650)
Inventories	1,320
Other current assets	1,850

Total current assets **\$11,940**

Current Liabilities

Accounts payable and other accrued liabilities	\$5,100
Current portion of borrowings	1,820
Other current liabilities	2,560

Total current liabilities **\$9,480**

The company's quick ratio is *closest* to:

- A 0.99.
- B 0.93.
- C 1.26.

B is correct.

Ratio	Formula	Calculation
Quick Ratio	(Cash + Marketable securities + Receivables)/Current liabilities	$(2,950 + 730 + 5,740 - 650)/9,480 = 0.93$

A is incorrect. It is the quick ratio calculated without removing the adjustment for doubtful accounts from accounts receivable.

Ratio	Formula	Calculation
Quick Ratio	(Cash + Marketable securities + Receivables)/Current liabilities	$(2,950 + 730 + 5,740 - 650)/9,480 = 0.93$
Quick Ratio with error	(Cash + Marketable securities + Receivables)/Current liabilities	$(2,950 + 730 + 5,740)/9,480 = 0.99$
Current Ratio	Current assets/Current liabilities	$11,940/9,480 = 1.3$

C is incorrect. It is the current ratio.

Ratio	Formula	Calculation
Quick Ratio	(Cash + Marketable securities + Receivables)/Current liabilities	$(2,950 + 730 + 5,740 - 650)/9,480 = 0.93$
Quick Ratio with error	(Cash + Marketable securities + Receivables)/Current liabilities	$(2,950 + 730 + 5,740)/9,480 = 0.99$
Current Ratio	Current assets/Current liabilities	$11,940/9,480 = 1.3$

Understanding Balance Sheets

- 55 Under US GAAP, interest paid is *most likely* included in which of the following cash flow activities?
- A Operating only
 - B Financing only
 - C Either operating or financing

A is correct. Interest paid must be categorized as an operating cash flow activity under US GAAP, although it can be categorized as either an operating or financing cash flow activity under IFRS.

B is incorrect. Interest paid cannot be categorized as a financing cash flow under US GAAP but can be under IFRS.

C is incorrect. Interest paid cannot be categorized as a financing cash flow under US GAAP but can be under IFRS.

Understanding Cash Flow Statements

- 56 Common-size financial statements are *most likely* a component of which step in the financial analysis framework?
- A Collect data
 - B Analyze/interpret data
 - C Process data



C is correct. Preparing common-size financial statements is part of the process data step.

A is incorrect. The financial statements are obtained in the collect data step, but not converted into common-size statements until the process step.

B is incorrect. Preparing common-size financial statements is part of the process data stage, after which the analyst will analyze/interpret the processed data.

Financial Statement Analysis: An Introduction

57 When using ratios to screen for potential equity investments, analysts can *most* effectively control for exposure to which of the following risks?

- A** Regulatory
- B** Financial
- C** Technological



B is correct. Ratio analysis provides a quick and direct way of comparing metrics addressing financial risk across a pool of companies.

A is incorrect. Regulatory risk is not readily assessed by ratio analysis. This risk is more likely to be assessed based on knowledge of regulatory trends, and the likely impact of regulatory change on individual companies.

C is incorrect. Technological risk is not readily assessed by ratio analysis. This risk is more likely to be assessed based on knowledge of technology trends in the market and the technological expertise of individual companies.

Financial Statement Analysis: Applications

58 Which of the following will be higher using the LIFO method compared with the FIFO method during periods of rising inventory unit costs?

- A** Gross profit
- B** Cost of sales
- C** Ending inventory



B is correct. Under either a perpetual or periodic inventory system, using the LIFO method will result in higher cost of sales than the FIFO method when inventory costs are increasing. This is because the cost allocated to cost of sales under the LIFO method more closely reflects current replacement values of inventory, which is higher than older inventory which was purchased at a lower cost.

A is incorrect because under either a perpetual or periodic inventory system, using the LIFO method will result in a higher cost of sales during periods of rising inventory costs because the cost of sales more closely reflects current replacement values. As such, the higher cost of sales will result in a lower gross profit, operating income, and net income as compared with the FIFO method.

C is incorrect because in an environment of rising inventory costs, ending inventory amounts under the LIFO method are typically not reflective of current replacement value because the ending inventory is assumed to be the oldest inventory, which is lower than current replacement costs. As such, ending inventory amounts under the LIFO method will be lower than the FIFO method.

Inventories

59 The following information is available for a company:

	2016 (in € millions)
EBIT (earnings before interest and taxes)	1,015.0
Interest expense	73.4
Tax expense	201.4
Total assets	5,305.0
Average total assets	5,421.0
Total debt	1,048.0
	2015
Interest coverage	15.3×
Debt to total assets	18.2%
Operating return on assets (ROA)	17.3%

Compared with 2015, which of the following ratios *most likely* indicates an improvement in the creditworthiness of the company? The change in the company's:

- A operating ROA.
- B debt-to-total assets.
- C interest coverage.

A is correct. When calculated for 2016, interest coverage decreased and debt to total assets and return on assets both increased. In general, a decrease in interest coverage and an increase in debt to total assets would reduce creditworthiness, but an increase in return on assets would improve creditworthiness. Calculations are as follows:

	Calculation	2016	Comment
Debt to total assets	Total debt/ Total assets = €1,048/€5,305 =	19.8%	Higher than 2015 (18.2%), which would reduce creditworthiness
Interest coverage	EBIT/Interest payments = €1,015/€73.4 =	13.8×	Lower than 2015 (15.3×), which would reduce creditworthiness
Operating ROA	EBIT/Average total assets = €1,015/€5,421 =	18.7%	Higher than 2015 (17.3%), which would improve creditworthiness

B is incorrect because an increase in debt-to-total assets would likely reduce creditworthiness.

C is incorrect because a decrease in the interest coverage ratio would likely reduce creditworthiness.

Financial Analysis Techniques

60 An income statement in a single-step format presents a subtotal for:

- A amortization of intangibles.
- B operating income.
- C gross margin.

B is correct. Single-step income statements present a subtotal for operating income.

A is incorrect because income statements in single-step format do not present a subtotal for amortization of intangibles.

C is incorrect because income statements in single-step format do not present a subtotal for gross margin. When an income statement shows a gross profit subtotal, it is said to use a multi-step format rather than a single-step format.

Understanding Income Statements

61 Which action is *most likely* considered a secondary source of liquidity?

- A Increasing the efficiency of cash flow management
- B Renegotiating current debt contracts to lower interest payments
- C Increasing the availability of bank lines of credit

B is correct. Renegotiating debt contracts is a secondary source of liquidity because it may affect the company's operating and/or financial positions.

A is incorrect. Increasing cash flow management efficiency is a primary source of liquidity.

C is incorrect. Increasing bank lines of credit is a primary source of liquidity.

Working Capital Management

62 Which of the following is *most likely* considered an example of matrix pricing when determining the cost of debt?

- A Debt-rating approach only.
- B Yield-to-maturity approach only.
- C Both the yield-to-maturity and the debt-rating approaches.

A is correct. The debt-rating approach is an example of matrix pricing.

B is incorrect because the yield-to-maturity approach is not an example of matrix pricing.

C is incorrect because the yield-to-maturity approach is not an example of matrix pricing.

Cost of Capital

Income Statement	Millions (\$)
Revenues	9.8
Variable operating costs	7.2
Fixed operating costs	1.5
Operating income	1.1
Interest	0.6
Taxable income	0.5
Tax	0.2
Net income	0.3

- 63 The degree of operating leverage (DOL) is *closest* to:
- A 2.4.
 - B 1.1.
 - C 1.7.

A is correct.

$$\begin{aligned}
 \text{DOL} &= \frac{\text{Revenues} - \text{Variable operating costs}}{\text{Revenues} - \text{Variable operating costs} - \text{Fixed operating costs}} \\
 &= \frac{9.8 - 7.2}{9.8 - 7.2 - 1.5} \\
 &= 2.36
 \end{aligned}$$

B is incorrect because it is Revenues/(Variable operating costs + Fixed operating costs).

C is incorrect because it is (Revenues – Variable operating costs)/Fixed operating costs.

Measures of Leverage

- 64 The acceptance of which of the following capital budgeting projects is *most likely* to expose a company to the highest level of uncertainty?
- A Replacement of worn out equipment
 - B Expansion projects
 - C Newly launched product or services

C is correct. Investments related to new products or services expose the company to even more uncertainties than expansion projects. These decisions are more complex and will involve more people in the decision-making process.

A is incorrect. Replacement of worn out equipment is simply an improvement to the existing project with recurring revenues.

B is incorrect. Investments related to new products or services expose the company to even more uncertainties than expansion projects. These decisions are more complex and will involve more people in the decision-making process

Capital Budgeting

- 65 The annual cost of trade credit assuming a 365-day year for terms 3/10 net 40 is *closest to*:
- A 43.3%.
 - B 44.9%.
 - C 32.0%.

B is correct.

$$\begin{aligned} &\text{Cost of trade credit} \\ &= \left[1 + \frac{\text{Discount}}{(1 - \text{Discount})} \right]^{(365/\text{Number of days beyond discount period})} - 1 \end{aligned}$$

$$\text{Cost of trade credit} = \left[1 + \frac{3\%}{(1 - 3\%)} \right]^{(365/30)} - 1 = 44.9\%$$

C is incorrect because the “number of days beyond discount” is set to 40.

A is incorrect because $(1 + 3\%)$ is the rate being compounded.

Working Capital Management

- 66 Which method of calculating the firm’s cost of equity is most likely to incorporate the long-run return relationship between the firm’s stock and the market portfolio?
- A Capital asset pricing model
 - B Dividend discount model
 - C Bond yield plus risk premium approach

A is correct. The capital asset pricing model uses the firm’s equity beta, which is computed from a market model regression of the company’s stock returns against market returns.

B is incorrect. Dividend discount model estimates the equity risk premium by adding the dividend yield and the growth rate in dividends.

C is incorrect. Cost of equity under this method is the additional of cost of debt and risk premium (the additional yield on a company’s stock relative to its bonds).

Cost of Capital

- 67 Financial risk is *least likely* affected by:
- A debentures.
 - B dividends.

- C long-term leases.

B is correct. By taking on fixed obligations, such as debt (including debentures) and long-term leases, a company increases its financial risk. Dividends will not increase financial risk.

A is incorrect because the use of debentures (one type of bond) is directly associated with financial risk.

C is incorrect because the use of long-term leases is directly related to financial risk.

Measures of Leverage

- 68 A firm with a marginal tax rate of 40% has a weighted average cost of capital of 7.11%. The before-tax cost of debt is 6%, and the cost of equity is 9%. The weight of equity in the firm's capital structure is *closest* to:

- A 79%.
B 65%.
C 37%.

B is correct.

$$\begin{aligned} \text{WACC} &= w_d r_d (1 - t) + w_e r_e, \text{ where } w_d + w_e = 1 \\ 7.11 &= (1 - w_e) \times 6 \times (1 - 0.4) + w_e \times 9 \\ w_e &= 65\% \end{aligned}$$

A is incorrect. It is calculated by dividing 7.11 by 9.

C is incorrect. It fails to adjust debt for taxes.

Cost of Capital

- 69 A company recently issued a 10-year, 6% semiannual coupon bond for \$864. The bond has a maturity value of \$1,000. If the marginal tax rate is 35%, the after-tax cost of debt (%) is *closest* to:

- A 3.9%.
B 5.2%.
C 2.6%.

B is correct. The pre-tax cost of debt is the yield to maturity (YTM) of the bond. The bond's YTM can be calculated by solving the following equation for i :

$$864 = \left(\sum_{t=1}^{20} \frac{30}{(1+i)^t} \right) + \frac{1000}{(1+i)^{20}}$$

Using a financial calculator, enter $N = 20$ (semiannual periods), $w = -864$, $\text{PMT} = 30$, and $\text{FV} = 1,000$. Compute I/YR . The six-month yield (or calculated I/YR) is 4%. The YTM is obtained by doubling the six-month yield to get 8%. Multiplying the pre-tax cost of debt by $(1 - \text{Tax rate})$ gives the result of $8 \times (1 - 0.35) = 5.2\%$.

A is incorrect. If the after-tax amount of the coupon rate is used, the result will be $0.06(1 - 0.35) = 3.9\%$.

C is incorrect. If the after-tax cost for six-month yield is used, the result will be $0.04(1 - 0.35) = 2.6\%$.

Cost of Capital

70 Based on good corporate governance practices, it is *most* appropriate for a company's compensation committee to:

- A** develop director remuneration policies.
- B** recommend remuneration for the external auditors.
- C** include some external directors.

A is correct. Under good corporate governance practices the compensation committee develops remuneration policies for directors as well as key executives. The audit committee, not the compensation committee, would be involved in the remuneration of the external auditors.

C is incorrect. The committee should be composed of independent (non-executive) members only.

B is incorrect. The audit committee is responsible for proposing the external auditor's remuneration.

Corporate Governance and ESG: An Introduction

71 A project has the following annual cash flows:

Year 0	Year 1	Year 2	Year 3
-\$606,061	\$2,151,515	-\$2,542,424	\$1,000,000

Which discount rate *most likely* provides a positive net present value (NPV)?

- A** 15%
- B** 21%
- C** 18%

B is correct.

Year	Cash Flow	K = 15%	K = 18%	K = 21%	Calculation
0	-\$606,061	-\$606,061.00	-\$606,061.00	-\$606,061.00	-606,061
1	2,151,515	1,870,882.60	1,823,317.80	1,778,111.60	$+2,151,515 / (1 + K)$
2	-2,542,424	-1,922,437.80	-1,825,929.30	-1,736,509.80	$-2,542,424 / (1 + K)^2$
3	1,000,000	657,516.20	608,630.90	564,473.90	$+1,000,000 / (1 + K)^3$

(continued)

Year	Cash Flow	K = 15%	K = 18%	K = 21%	Calculation
	NPV	-\$100.00	-\$41.60	+\$14.70	K = Discount rate

The NPV at 21% is \$14.7, whereas the other two NPVs are negative.

A is incorrect because it has a negative NPV.

C is incorrect because it has a negative NPV.

Capital Budgeting

72 Which of the following is *least likely* to be of concern to values-based ESG investors?

- A** Avoidance of companies that conflict with moral values
- B** Reduction in risks associated with increased litigation costs
- C** Increase in risk-adjusted returns through ESG factor ranking

A is correct. The objective of a values-based ESG approach is to mitigate risks and identify opportunities by analyzing ESG considerations in addition to traditional finance metrics. Avoidance of companies that conflict with moral or ethical values reflects a values-based approach.

B and C are incorrect. The objective of a values-based ESG approach is to mitigate risks and identify opportunities by analyzing ESG considerations in addition to traditional finance metrics.

Corporate Governance and ESG: An Introduction

73 The behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains is *best* described as:

- A** mental accounting.
- B** the gambler's fallacy.
- C** the disposition effect.

C is correct. Behavioral biases in which investors tend to avoid realizing losses but rather seek to realize gains is the disposition effect.

A is incorrect. The disposition effect is a behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains.

B is incorrect. The gambler's fallacy is a behavioral bias in which recent outcomes affect investors' estimates of future probabilities.

Market Efficiency

74 Returns from a depository receipt are *least likely* affected by which of the following factors?

- A** Exchange rate movements

- B Number of depository receipts
- C Analysts' recommendations

B is correct. The price of each depository receipt (and, in turn, returns) will be affected by factors that affect the price of the underlying shares—such as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements. The number of depository receipts issued affects their price but does not affect the returns.

A is incorrect. The price of each depository receipt (and, in turn, returns) will be affected by such factors as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements.

C is incorrect. The price of each depository receipt (and, in turn, returns) will be affected by such factors as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements.

Overview of Equity Securities

75 Which of the following statements is *least* accurate with respect to fixed-income indexes?

- A Many of the underlying securities in the index tend to be illiquid.
- B The indexes are susceptible to turnover of the underlying securities.
- C Compared with equity indexes, it is easier and less expensive to replicate fixed-income indexes.

C is correct. Compared with equity indexes, the large number of fixed-income securities—combined with the lack of liquidity of some securities—has made it more costly and difficult for investors to replicate fixed-income indexes and duplicate their performance.

A is incorrect. Many of the underlying securities in the index tend to be illiquid because fixed-income markets are predominantly dealer markets.

B is incorrect. The indexes are susceptible to turnover of the underlying securities because over time, fixed-income securities mature and issuers offer new securities to meet their financial needs.

Security Market Indexes

76 Which of the following is *most likely* one of the main functions of the financial system?

- A Determining an equilibrium interest rate
- B Ensuring that markets are informationally efficient
- C Ensuring that all investment projects receive sufficient funding

A is correct. One of the main functions of the financial system is to determine the equilibrium interest rate, which is the only interest rate that would exist if all securities were equally risky, had equal terms, and were equally liquid.

B is incorrect. Informational market efficiency is not a key function of the financial system, rather that of regulatory framework

C is incorrect. The financial system provides sufficient funding only to the most productive projects. An important function of the financial system is to direct resources away from wealth-diminishing projects.

Market Organization and Structure

77 An equity analyst follows two industries with the following characteristics:

Industry 1:

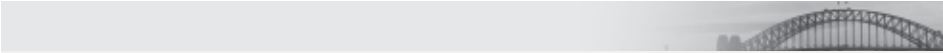
A few companies with proprietary technologies, products with unique features, high switching costs, and minimal regulatory influences.

Industry 2:

A few companies producing relatively similar products, sales varying with disposable income and employment levels, high capital costs and investment in physical plants, rapid shifts in market shares of competing firms, and minimal regulatory influences.

Based on the above information, the analyst will *most* appropriately conclude that, compared with the firms in Industry 2, those in Industry 1 would potentially have:

- A** over-capacity problems.
- B** high bargaining power of customers.
- C** larger economic profits.



C is correct. The economic profit (the spread between the return on invested capital and the cost of capital) tends to be larger in industries with differentiated products, greater pricing power, and high switching costs to customers. Industry 1 has these features. In contrast, firms in Industry 2 have little pricing power (undifferentiated products and rapid shifts in market shares, indicating intense rivalry), which is indicative of potentially smaller economic profits.

A is incorrect. The characteristics of Industry 1 do not indicate a potential for over-capacity problems. If anything, Industry 2 is prone to such a problem because of high capital costs and investment in physical capital, cyclical demand for products, and rapid shifts in market shares.

B is incorrect. Industry 1 is less prone to the bargaining power of customers because of differentiated products and high switching costs for customers.

Introduction to Industry and Company Analysis

78 For a US investor, which of the following statements concerning investing in depository receipts (DRs) is *least* accurate?

- A** Investing in DRs could provide arbitrage opportunities and entail currency risk.
- B** Investors in unsponsored DRs would have the same voting rights as the direct owners of common shares.
- C** Sponsored DRs are subject to greater reporting requirements than unsponsored DRs.



B is correct. Investors of unsponsored DRs would not have the same voting rights as the direct owners of common shares because the depository bank retains the voting rights.

A is incorrect because it is an accurate statement. The DRs trading on multiple exchanges could experience short-term valuation discrepancies, potentially giving rise to a quick arbitrage profit opportunity for astute traders to exploit. The price of each DR will be affected by factors that affect the price of the underlying shares and exchange rate movements.

C is incorrect because it is an accurate statement. Sponsored DRs are subject to greater reporting requirements than unsponsored DRs. In the United States, sponsored DRs must be registered with the SEC.

Overview of Equity Securities

79 An industry experiencing intense competitive rivalry among incumbent companies is *best* characterized by:

- A** differentiated products and low exit barriers.
- B** a small number of competitors and low fixed costs.
- C** customers basing purchase decisions largely on price.



C is correct. The factor that most influences customer purchase decisions is likely to also be the focus of competitive rivalry in the industry. In general, industries in which price is a large factor in customer purchase decisions tend to be more competitive than industries in which customers value other attributes more highly.

A is incorrect. Industries experiencing more intense rivalry among incumbent companies are characterized by undifferentiated products and high exit barriers.

B is incorrect. Industries experiencing more intense rivalry among incumbent companies are fragmented among many small competitors and they tend to have high fixed costs.

Introduction to Industry and Company Analysis

80 Which of the following is *most* likely associated with secondary capital markets?

- A** Continuous trading
- B** Lead underwriters
- C** Book building



A is correct. A secondary market is where investors continue to trade the securities among themselves. Continuous trading is typically associated with secondary capital markets.

B is incorrect. Lead underwriter is associated with primary markets.

C is incorrect. Book building is associated with primary markets.

Market Organization and Structure

81 An investor gathers the following information about a company:

Current dividend per share	\$3
Historical annual dividend growth rate	4%
Expected annual dividend growth rate for the next three years	8%
Expected stock value per share at the end of Year 3	\$33

If the investors' required rate of return is 15%, the current estimate of the intrinsic value per share is *closest* to:

- A \$28.36.
- B \$29.65.
- C \$29.08.

B is correct.

$$\begin{aligned}
 V_0 &= \frac{3 \times 1.08}{1 + 0.15} + \frac{3 \times (1.08)^2}{(1 + 0.15)^2} + \frac{[3 \times (1.08)^3] + 33}{(1 + 0.15)^3} \\
 &= 2.82 + 2.65 + 2.48 + 21.70 \\
 &= \$29.65
 \end{aligned}$$

A is incorrect. It uses the historical growth rate and the constant growth model for estimating the intrinsic value.

$$\begin{aligned}
 V_0 &= \frac{3 \times 1.04}{0.15 - 0.04} \\
 &= \$28.36
 \end{aligned}$$

B is incorrect. It uses the historical growth rate rather than an analyst's growth forecast.

$$\begin{aligned}
 V_0 &= \frac{3 \times 1.04}{1 + 0.15} + \frac{3 \times (1.04)^2}{(1 + 0.15)^2} + \frac{[3 \times (1.04)^3] + 33}{(1 + 0.15)^3} \\
 &= 2.71 + 2.45 + 2.22 + 21.70 \\
 &= \$29.08
 \end{aligned}$$

Equity Valuation: Concepts and Basic Tools

82 An equity index consists of three securities with market information as follows:

Security	Shares Outstanding	Price at Beginning of Period	Price at End of Period	Dividend per Share
A	5,000,000	\$10.00	\$9.50	\$1.00
B	2,000,000	\$20.00	\$21.50	\$0.80
C	1,500,000	\$30.00	\$33.00	\$0.60

The price-weighted total return index is *closest* to:

- A 10.7%.
- B 6.7%.
- C 9.5%.

A is correct.

Calculation of Index and Return Based on Price-Weighted Method

Security	Shares Outstanding	BOP Price	EOP Price	Dividends per Share
A	5,000,000	\$10.00	\$9.50	\$1.00
B	2,000,000	\$20.00	\$21.50	\$0.80
C	1,500,000	\$30.00	\$33.00	\$0.60
Total		\$60.00	\$64.00	\$2.40
Index value		20.00	21.33	0.80

Type of Index	BOP Value	EOP Value	Return
Price return	\$20.00	\$21.33	6.65%
Total return	\$20.00	\$22.13	10.65%

BOP = Beginning of period; EOP = Ending of period

B is incorrect. It is the price-weighted index based on price changes only; it ignores dividends.

C is incorrect. It is the return based on equal-weighted method.

Calculation of Index and Return Based on Equal-Weighted Method

Security	BOP Price	EOP Price	Dividends per share	Price Return (%)	Total Return (%)	Weight (33.3%) × Price Return (%)	Weight (33.3%) × Total Return (%)
A	10.00	9.50	1.00	-5.00%	5.00%	-1.67%	1.67%
B	20.00	21.50	0.80	7.50%	11.50%	2.50%	3.83%
C	30.00	33.00	0.60	10.00%	12.00%	3.33%	4.00%
Total						4.17%	9.50%

BOP = Beginning of period; EOP = Ending of period

Security Market Indexes

83 A company has issued non-callable, non-convertible preferred stock with the following features:

- Par value per share: \$10
- Annual dividend per share: \$2
- Maturity: 15 years

An investor's required rate of return is 8%, and the current market price per share of the preferred stock is \$25. By comparing the estimated intrinsic value with the market price of the preferred stock, the *most likely* conclusion is that the preferred stock is:

- A fairly valued at \$25.00.
- B overvalued by \$4.73.
- C undervalued by \$15.00.

B is correct. Using a financial calculator to find the present value (PV) of the future cash flows, intrinsic value is thus: $FV = \$10$; $n = 15$; $PMT = \$2$; $r = 8\%$. Compute $PV = \$20.27$.

The preferred stock is overvalued by \$4.73 (Market price of \$25 – Estimated value of \$20.27).

A is incorrect. It ignores the maturity information and treats it as a perpetual preferred stock.

$$V_0 = D_0/r = \$2/0.08 = \$25$$

C is incorrect. It ignores the time value of money and simply sums up the total of dividends and the par value.

$$V_0 = (\text{Annual dividend per share} \times \text{Maturity}) + \text{Par Value} = (2 \times 15) + 10 = 40$$

Thus, the preferred stock appears to be undervalued by \$15.00 (Market price of \$25 – Estimated value of \$40).

Equity Valuation: Concepts and Basic Tools

84 An analyst collects the following data on a company:

Return on Equity (%)	Payout Ratio (%)	Justified Forward P/E
12.0	30	7.5

Using a required return of 12.4%, if the company increases its dividend payout ratio to 40%, the justified forward P/E ratio will be *closest* to:

- A 11.5.
- B 7.7.
- C 5.3.

B is correct.

$$\text{Dividend growth rate} = (1 - \text{Payout ratio}) \times \text{ROE}$$

$$\text{Justified forward P/E: } P_0/E_1 = p/(r - g), \text{ where } p = \text{Payout ratio.}$$

Using the new payout ratios, the justified forward P/E ratios, are calculated:

$$\text{New dividend growth rate} = (1 - 0.4) \times 12\% = 7.2\%;$$

$$\text{New justified forward P/E} = 0.4/(0.124 - 0.072) = 7.7\times.$$

C is incorrect. It uses wrong dividend growth rate by taking a product of payout ratio and ROE.

$$\text{New dividend growth rate} = (0.4) \times 12\% = 4.8\%;$$

$$\text{New Justified forward P/E} = 0.4 / (0.124 - 0.048) = 5.3\times$$

A is incorrect. It uses payout ratio instead of retention rate to calculate new justified P/E ratio.

$$\text{New dividend growth rate} = (1 - 0.4) \times 12\% = 7.2\%;$$

$$\text{New justified forward P/E} = (1 - 0.4) / (0.124 - 0.072) = 11.5\times$$

Equity Valuation: Concepts and Basic Tools

- 85 The value effect market-pricing anomaly *most likely* occurs when stocks that have below-average price-to-earnings and market-to-book ratios, as well as above-average dividend yields, consistently outperform:

- A large-cap stocks.
- B growth stocks.
- C stocks that have had negative earnings surprises.



B is correct. The value effect occurs when value stocks, which are generally referred to as stocks that have below-average price-to-earnings and market-to-book ratios, as well as above-average dividend yields, outperform growth stocks consistently and for long periods.

A is incorrect. It is the size effect, not the value effect, that compares returns with respect to large-cap stocks.

C is incorrect. Value effect does not compare stocks on the basis of earnings surprises. It is the "earnings surprise" anomaly that compares companies on the basis of the unexpected part of the earnings announcement.

Market Efficiency

- 86 Ted Nguyen is an investor domiciled in a country with an original issue discount tax provision. He purchases a zero-coupon bond at a deep discount to par value with the intention of holding the bond until maturity. At maturity, he will *most likely* face:

- A a capital gain.
- B neither a capital loss nor gain.
- C a capital loss.



B is correct. An original issue discount tax provision allows the investor to increase the cost basis of the bond, so when the bond matures, the investor faces no capital gain or loss.

A is incorrect because the investor will face neither a capital gain nor a loss as the original issue discount tax provision allows the investor to increase the cost basis of the bond.

C is incorrect because the investor will face neither a capital gain nor a loss as the original issue discount tax provision allows the investor to increase the cost basis of the bond.

Fixed-Income Securities: Defining Elements

- 87** Which of the following is *least likely* a short-term funding method available to banks?
- A** Central bank funds
 - B** Negotiable certificate of deposits
 - C** Syndicated loans

C is correct. A syndicated loan is a loan from a group of lenders, called the “syndicate,” to a single borrower. Syndicated loans are primarily originated by banks, and the loans are extended to companies but also to governments and government-related entities.

A is incorrect because central bank funds are one of the short-term wholesale funds available to banks for short-term funding needs.

B is incorrect because a negotiable CD allows any depositor (initial or subsequent) to sell the CD in the open market prior to the maturity date. CDs are an important source of funds for financial institutions.

Fixed-Income Markets: Issuance, Trading, and Funding

- 88** In a securitization structure, credit tranching allows investors to choose between:
- A** subordinated bonds and senior bonds.
 - B** extension risk and contraction risk.
 - C** partially amortizing loans and fully amortizing loans.

A is correct. Credit tranching allows investors to choose between subordinate and senior bond classes as a means of credit enhancement. The purpose of this structure is to redistribute the credit risk associated with the collateral.

B is incorrect because extension and contraction risks refer to the types of prepayment risks in a securitization structure.

C is incorrect because partially and fully amortizing loans refer to two ways in which the loan principal can be repaid.

Introduction to Asset-Backed Securities

- 89** DMT Corp. issued a five-year floating-rate note (FRN) that pays a quarterly coupon of three-month Libor plus 125 bps. The FRN is priced at 96 per 100 of par value. Assuming a 30/360-day count convention, evenly spaced periods, and constant three-month Libor of 5%, the discount margin for the FRN is *closest* to:
- A** 180 bps.
 - B** 400 bps.
 - C** 221 bps.

C is correct. The interest payment each period per 100 of par value is

$$\frac{(\text{Index} + QM) \times FV}{m} = \frac{(0.05 + 0.0125) \times 100}{4} = 1.5625$$

The discount margin can be estimated by solving for DM in the equation

$$96 = \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^1} + \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^2} + \dots + \frac{1.5625}{\left(1 + \frac{0.05 + DM}{4}\right)^{20}}$$

The solution for the discount rate, $r = (0.05 + DM)/4$ is 1.8025%. Therefore, $DM = 2.21\%$ or 221 bps.

A is incorrect because it uses the quarterly discount rate 1.8025% as the solution, i.e., 180 bps.

B is incorrect because it uses the price discount of $100\% - 96\% = 4\%$, i.e., 400 bps.

Introduction to Fixed-Income Valuation

- 90 The bonds of Apex Corporations have a par value of \$10,000 each and an annual required rate of return of 10%. The bonds make quarterly coupon payments at an annual rate of 6% and have two years remaining until maturity. The current market price of each bond is *closest to*:

- A \$10,749.
- B \$9,283.
- C \$9,306.

B is correct. Using the quarterly coupon payment of \$150 [= $(0.06 \times 10000)/4$] over eight quarters and a quarterly required rate of return of 2.5%, we calculate the bond's price as:

$$P_0 = 150/(1.025)^1 + 150/(1.025)^2 + \dots + 150/(1.025)^8 + 10,000/(1.025)^8 \\ = \$9,282.99$$

A is incorrect because the bond's price is computed using a quarterly coupon payment of \$250 [= $(0.10 \times 10000)/4$] and a quarterly required rate of return of 1.5%:

$$P_0 = 250/(1.015)^1 + 250/(1.015)^2 + \dots + 250/(1.015)^8 + 10000/(1.015)^8 \\ = \$10,748.59$$

C is incorrect because the bond's price is computed using the annual coupon payment of \$600 [= $(0.06 \times 10,000)$] over 2 years and the annual required rate of return of 10%:

$$P_0 = 600/(1.10)^1 + (600 + 10,000)/(1.10)^2 \\ = \$9,305.79$$

Introduction to Fixed-Income Valuation

- 91 A bond's duration is 7.31, and its convexity is -24.85. Using the duration model with convexity adjustment, the bond's percentage change in price if interest rates decrease 2% is *closest to*:

- A 15.12%.

B 15.60%.

C 14.12%.

C is correct. The duration model estimates the percentage change in price as $-\text{AnnModDur} \times \Delta\text{Yield}$, or $-7.31 \times (-0.02) = +14.62\%$, and the convexity adjustment is $\frac{1}{2} \times \text{AnnConvexity} \times (\Delta\text{Yield})^2$, or $\frac{1}{2} \times (-24.85) \times (0.02)^2 = -0.50\%$, and $14.62\% - 0.50\% = 14.12\%$.

A is incorrect because it uses $-C \times \Delta i$ as the convexity adjustment or $-24.85 \times -0.02 = 0.50\%$ and $14.62\% + 0.50\% = 15.12\%$.

B is incorrect because it uses $-C \times (\Delta i)^2 \times 100$ or $-24.85 \times (0.02)^2 \times 100 = +0.98\%$ as the convexity adjustment and $14.62\% + 0.98\% = 15.60\%$.

Understanding Fixed-Income Risk and Return

- 92** Consider the following information on three bonds: Bond A, Bond B, and Bond C.

	Bond A	Bond B	Bond C
Time to maturity (years)	10	6	4
Par value	\$10,000,000	\$10,000,000	\$10,000,000
Market value	\$9,323,381	\$11,189,092	\$10,000,000
Annualized modified duration	7.3105	4.6880	3.4370

A fixed-income portfolio contains Bond A and Bond B. If another \$10 million is added to the portfolio and is used to purchase Bond C, then the annualized modified duration of the portfolio will be *closest* to:

A 5.0793.

B 5.1452.

C 6.5668.

A is correct. The new modified duration of the portfolio is calculated as follows:

The initial market value of the portfolio = $\$9,323,381 + \$11,189,092 = \$20,512,473$.

The portfolio market value with the purchase of Bond C is:

New portfolio market value = $\$20,512,473 + \$10,000,000 = \$30,512,473$.

$$\begin{aligned} \text{ModDur} &= \left[7.3105 \times \frac{9,323,381}{30,512,473} \right] + \left[4.6880 \times \frac{11,189,092}{30,512,473} \right] + \left[3.4370 \times \frac{10,000,000}{30,512,473} \right] \\ &= 5.0793 \end{aligned}$$

where the first term in each set of brackets is the annualized modified duration and the second term is the weighting factor based on market value.

B is incorrect because the calculation of 5.1452 uses par value rather than market value in determining the portfolio value of \$30,000,000 and in setting the weighting factors:

$$\begin{aligned} \text{ModDur} &= \left[7.3105 \times \frac{10,000,000}{30,000,000} \right] + \left[4.6880 \times \frac{10,000,000}{30,000,000} \right] + \left[3.4370 \times \frac{10,000,000}{30,000,000} \right] \\ &= 5.1452 \end{aligned}$$

C is incorrect because the calculation of 6.5668 uses the time to maturity rather than the annualized modified duration for each bond:

$$\text{ModDur} = \left[10 \times \frac{9,323,381}{30,512,473} \right] + \left[6 \times \frac{11,189,092}{30,512,473} \right] + \left[4 \times \frac{10,000,000}{30,512,473} \right]$$

$$= 6.5668$$

Understanding Fixed-Income Risk and Return

93 The estimated percentage change in bond price for a given change in the term structure of yield volatility depends on modified duration and:

- A convexity only.
- B the change in yield to maturity only.
- C both convexity and the change in yield to maturity.

C is correct. The estimated percentage change in the bond price depends on modified duration and convexity as well as on the yield-to-maturity change.

A is incorrect because the estimated percentage change in the bond price depends on the change in yield to maturity as well as modified duration and convexity.

B is incorrect because the estimated percentage change in the bond price depends on convexity as well as modified duration and the change in yield to maturity.

Understanding Fixed-Income Risk and Return

94 The yield spread on a corporate bond would *most likely* narrow as:

- A issuer size increases.
- B downgrade risk increases.
- C trading frequency decreases.

A is correct. Market liquidity risk is affected by the size of the issuer and the credit quality of the issuer. In general, the less debt an issuer has outstanding, the less frequently it trades, the higher the market liquidity risk. Thus, an increase in issue size (publicly traded debt the issuer has outstanding) is consistent with less market liquidity risk and narrower spreads.

B is incorrect. Downgrade risk refers to the risk that a bond issuer's creditworthiness deteriorates, leading investors to believe the risk of default is higher and thus causing the yield spreads on the issuer's bonds to widen.

C is incorrect. As trading frequency decreases, market liquidity risk increases. An increase in market liquidity is consistent with yield spreads widening.

Fundamentals of Credit Analysis

95 Below are several key statistics for three countries:

(In billions except where noted)	Country A	Country B	Country C
Nominal GDP	\$1,796.19	\$1,511.25	\$53.24
Population (millions)	206.10	50.80	7.13
Government expenditures	\$749.00	\$338.00	\$20.31
Government revenues	\$726.60	\$351.60	\$19.53
Current account balance	−\$23.53	\$98.68	\$2.20
Imports	\$139.40	\$371.10	\$27.00
Exports	\$184.50	\$511.80	\$24.96
Government debt	\$1,329.90	\$544.74	\$15.44

Based on the statistics provided, which country is *most likely* to be the highest-rated credit?

- A Country A
- B Country B
- C Country C

B is correct. Overall, Country B will most likely have the highest credit rating. Country B has the highest GDP per capita, the highest current account as a share of GDP, the highest budget and trade surpluses, and a relatively low government debt-to-GDP ratio (second only to Country C).

	Country A	Country B	Country C
Government debt/GDP	74.04%	36.05%	29.00%
GDP per capita	\$8,715.14	\$29,749.02	\$7,467.04
Current account as share of GDP (%)	−1.31%	6.53%	4.13%
Budget balance (surplus/deficit)	−\$22.40	\$13.60	−\$0.78
Balance of trade (surplus/deficit)	\$45.10	\$140.70	−\$2.04

A is incorrect. Country A has the highest government debt/GDP of the three countries and a relatively low GDP per capita. Additionally, its current account as a share of GDP is negative, which means it must attract funding from outside the country. On the bright side, the country has a trade surplus, but that is offset by the budget deficit and other factors already mentioned.

C is incorrect. Country C does have the lowest government debt/GDP, but only by a small margin. In every other category, Country B appears stronger than Country C.

Fundamentals of Credit Analysis

- 96 A key distinction between commercial mortgage-backed securities (CMBSs) and residential mortgage-backed securities (RMBSs) is that CMBSs *most likely*:
- A have balloon maturities.
 - B have amortizing principal.
 - C do not have call protection.



A is correct. Many commercial loans backing CMBs are balloon loans that require a substantial principal repayment at maturity of the loan. If the borrower fails to make the balloon payment, the borrower is in default. CMBs generally do not have amortizing principal as RMBs do, and they do offer call protection.

B is incorrect. CMBs generally do not have amortizing principal as RMBs do.

C is incorrect. CMBs generally do offer call protection.

Introduction to Asset-Backed Securities

97 An inverse floater will *most likely* have:

- A a maximum coupon rate.
- B a face value that changes as the reference rate changes.
- C a coupon rate that changes by more than the change in the reference rate.



A is correct. The general formula for the coupon rate of an inverse floater is $C - (L \times R)$, where C is the maximum coupon rate if the reference rate (R) is equal to zero and L is the coupon leverage, which is greater than zero.

B is incorrect. The face value of an inverse floater does not change as the reference rate changes; rather, the coupon rate changes.

C is incorrect. Inverse floaters can have leverage less than 1 (deleveraged inverse floaters), so the coupon rate changes by less than the change in the reference rate; leverage greater than 1 (leveraged inverse floaters), so the coupon rate changes by more than the change in the reference rate; or leverage equal to 1, so the coupon rate changes by the same amount as the change in the reference rate.

Fixed-Income Markets: Issuance, Trading, and Funding

98 An analyst gathers the following information on a bond:

Approximate modified duration	10.3829
Approximate modified convexity	141.217

What is the estimated percentage price change for the bond if yield to maturity decreases by 50 bps?

- A -5.01%
- B 5.37%
- C 5.54%

B is correct. The convexity-adjusted estimate of the percentage change in bond price is calculated as follows:

$$\begin{aligned}\% \Delta PV^{Full} &\approx (-\text{AnnModDur} \times \Delta \text{Yield}) + \left[\frac{1}{2} \times \text{AnnConvexity} \times (\Delta \text{Yield})^2 \right] \\ &\approx (-10.3829 \times -0.005) + \left[\frac{1}{2} \times 141.217 \times (-0.005)^2 \right] \\ &= 0.051915 + 0.001765 \\ &= 0.05368\end{aligned}$$

C is incorrect. It is the result of neglecting to multiply the convexity adjustment by ½.

$$\begin{aligned}&\approx (-10.3829 \times -0.005) + \left[141.217 \times (-0.005)^2 \right] \\ &= 0.051915 + 0.00353 \\ &= 0.0554\end{aligned}$$

A is incorrect. It is the result of omitting the negative sign for the change in yield.

$$\begin{aligned}&\approx (-10.3829 \times 0.005) + \left[\frac{1}{2} \times 141.217 \times (-0.005)^2 \right] \\ &= -0.051915 + 0.001765 \\ &= -0.0501\end{aligned}$$

Understanding Fixed-Income Risk and Return

- 99 A swap in which the investor receives a variable payment in line with market conditions and makes a fixed payment can *best* be replicated by purchasing a:
- A set of long futures contracts which are matched with short forward contracts.
 - B series of forward contracts, each with an initial value of zero.
 - C floating rate bond financed using a fixed-rate bond.

C is correct. The payment structure is replicated by being long the floating rate bond and being short the fixed-rate bond.

A is incorrect. This strategy does not replicate a swap in which the investor receives a variable payment in line with market conditions and makes a fixed payment.

B is incorrect. Due to differences in timing, the forward contracts need to be off-market contracts.

Basics of Derivative Pricing and Valuation

- 100 Which of the following is *least likely* to be an example of a derivative?
- A An exchange-traded fund
 - B A contract to sell Alphabet Inc.'s shares at a fixed price
 - C A contract to buy Australian dollars at a predetermined exchange rate



A is correct. Although an exchange-traded fund derives its value from the underlying assets it holds, it does not transform the performance of those assets and so is not a derivative.

B is incorrect. A contract to sell Alphabet Inc.'s shares transforms the performance of the underlying shares of Alphabet Inc and is an example of an option derivative.

C is incorrect. A contract to buy Australian dollars transforms the performance of the underlying currency and is an example of a currency derivative.

Derivative Markets and Instruments

101 Which of the following derivatives is *least likely* to be classified as a contingent claim?

- A** A futures contract
- B** A call option contract
- C** A credit default swap



A is correct. A futures contract is classified as a forward commitment in which the buyer undertakes to purchase the underlying asset from the seller at a later date and at a price agreed on by the two parties when the contract is initiated.

B is incorrect. A call option contract is a contingent claim in which the buyer of the option has a right to purchase the underlying asset at a fixed price on or before a pre-specified expiration date.

C is incorrect. A credit default swap is a contingent claim in which the credit protection seller provides protection to the credit protection buyer against the credit risk of a third party.

Derivative Markets and Instruments

102 In a credit default swap, the party that receives a series of cash payments in return for promising to pay compensation for credit losses resulting from a third party's default is *most likely* the:

- A** clearinghouse.
- B** seller of the swap.
- C** buyer of the swap.



B is correct. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

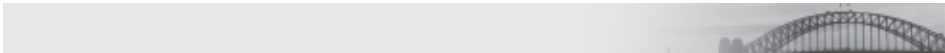
A is incorrect. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller (not the clearinghouse) receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

C is incorrect. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller (not the buyer) receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

Derivative Markets and Instruments

103 All, else held equal, the value of a European call option is *best* characterized as having a:

- A** negative relationship with the price of the underlying.
- B** negative relationship with the volatility of the underlying.
- C** positive relationship with the time to expiration.



C is correct. The value of a European call option is directly related to the time to expiration. That is, all else held equal, the value of a European call option is higher the longer the time to expiration.

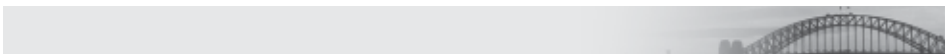
A is incorrect. The value of a European call option is directly related to the price of the underlying.

B is incorrect. The value of a European call option is directly related to the volatility of the underlying.

Basics of Derivative Pricing and Valuation

104 If dividends paid by the underlying increase, the value of a European call option will *most likely*:

- A** not change.
- B** increase.
- C** decrease.



C is correct. A European call option is worth less the more dividends are paid by the underlying.

A is incorrect. A European call option is worth less the more dividends are paid by the underlying.

B is incorrect. A European call option is worth less the more dividends are paid by the underlying.

Basics of Derivative Pricing and Valuation

105 For an investor with a long position, the price of a futures contract will *most likely* be higher than the price on a forward contract on the same asset with the same expiration date if there is a:

- A** zero correlation between the futures price and interest rates.
- B** positive correlation between the futures price and interest rates.
- C** negative correlation between the futures price and interest rates.



B is correct. If there is a positive correlation between futures prices and interest rates, an investor with a long position will favor futures over forwards because rising prices lead to futures profits that are reinvested at higher interest rates. With forwards, all the gains are received at expiration and thus there is no gain from reinvestment.

A is incorrect. In this case, futures and forward prices will be the same.

C is incorrect. With negative correlation between futures prices and interest rates, falling prices lead to losses during periods of rising interest rates. Forward prices will be higher than futures prices.

Basics of Derivative Pricing and Valuation

106 The real estate index *most likely* to suffer from sample selection bias is a(n):

- A** repeat sales index.
- B** REIT index.
- C** appraisal index.



A is correct. Only properties that sell in each period and are included in the index and vary over time which may not be representative of the whole market.

B is incorrect. The REIT index is based on a set of publicly traded REITs and thus does not suffer from sample selection bias.

C is incorrect. The appraisal index is based on a set of properties that is appraised regularly. Thus it does not suffer from sample selection bias.

Introduction to Alternative Investments

107 In the context of venture capital financing, seed-stage financing *most likely* supports:

- A** initial commercial production and sales.
- B** product development and/or marketing efforts.
- C** transformation of an idea into a business plan.



B is correct. Support of product development and/or marketing efforts takes place during seed-stage financing.

A is incorrect. Support of initial commercial production and sales takes place during early stage financing.

C is incorrect. Support in the transformation of an idea into a business plan takes place during angel investing.

Introduction to Alternative Investments

108 Relative to traditional investments, alternative investments are *best* characterized as having:

- A** higher correlations with other asset classes.
- B** unique legal and tax considerations.

- C greater liquidity.



B is correct. Alternative investments are more likely characterized as having unique legal and tax considerations because of the broad range and complexity of the investments.

A is incorrect because alternative investments typically have lower correlations with traditional investments, such as stocks, bonds, and cash.

C is incorrect because alternative investments typically have underlying investments that are illiquid.

Introduction to Alternative Investments

- 109 For a hedge fund investor, a benefit of investing in a fund of funds is *least likely* the:

- A higher level of due diligence expertise.
B multilayered fee structure.
C ability to negotiate better redemption terms.



B is correct. Funds of funds have a multilayered fee structure that will reduce the returns to the investor.

A is incorrect because one advantage of fund of funds is that they usually have a high level of due diligence expertise.

C is incorrect because another advantage of fund of funds is their ability to negotiate better redemption terms such as shorter lockup and notice periods.

Introduction to Alternative Investments

- 110 If the level of broad inflation indexes is largely determined by commodity prices, the average real yield on direct commodity investments is *most likely*:

- A less than zero.
B equal to zero.
C greater than zero.



B is correct. As the price increases of commodities are mirrored in higher price indexes, the nominal return is equal to inflation and thus the real return is zero.

A is incorrect. A negative real return implies inflation greater than nominal return. This is not the case if commodity price increases determine inflation.

C is incorrect. A positive real return implies inflation less than nominal return. This is not the case if commodity price increases determine inflation.

Introduction to Alternative Investments

- 111 An investor seeking an indirect debt investment in real estate will:

- A purchase a mortgage-backed security.

- B purchase a commercial property.
- C originate a residential mortgage.

A is correct. Mortgage-backed securities are pools of loans that are securitized and offered to the financial markets providing indirect debt investment opportunities in residential property.

C is incorrect because originators (generally financial institutions) of residential mortgages are making a direct debt investment in the home.

B is incorrect because commercial property is considered an appropriate direct investment—equity (i.e., ownership) and debt (i.e. lender)—for institutional or high-net-worth investors with long time horizons and limited liquidity needs.

Introduction to Alternative Investments

- 112 A hedge fund with \$98 million of initial capital charges a management fee of 2% and an incentive fee of 20%. The management fee is based on assets under management at year end and the incentive fee is calculated independently from the management fee. The fee structure has a high-water mark provision. The fund value is \$112 million at the end of Year 1, \$100 million at the end of Year 2, and \$116 million at the end of Year 3. The net-of-fees return earned by the fund in Year 3 is *closest* to:

- A 14.15%.
- B 12.33%.
- C 11.87%.

A is correct. The net-of-fees return to the fund in Year 3 is closest to 14.15%, calculated as follows:

Year 1:

Portfolio gain = Year-end value – Beginning value = \$112 million – \$98 million = \$14 million

Management fee = Year-end value × Management fee % =
\$112 million × 2% = \$2.24 million

Incentive fee = Portfolio gain × Incentive fee % = \$14 million ×
20% = \$2.8 million

Total fees = Management fee + Incentive fee = \$2.24 million + \$2.8 million = \$5.04 million

Ending Capital Position = Year-end value – Total fees
= \$112 million – \$5.04 million = \$106.96 million

High water mark = Ending capital position = \$106.96 million

Year 2:

No incentive fee is earned as the fund declines in value; the high water mark established in Year 1 is not exceeded.

$$\begin{aligned}\text{Management fee} &= \text{Year-end value} \times \text{Management fee \%} = \\ &\quad \$100 \text{ million} \times 2\% = \$2 \text{ million} \\ \text{Ending Capital Position} &= \text{Year-end value} - \text{Management fee} \\ &= \$100 \text{ million} - \$2 \text{ million} = \$98 \text{ million} \\ \text{High water mark} &= \text{Highest ending capital position} = \\ &\quad \$106.96 \text{ million}\end{aligned}$$

Year 3:

Net-of-fee returns are affected by the Year 1 high water mark and the Year 2 net capital position (i.e. Year 3 beginning capital position).

$$\begin{aligned}\text{Management fee} &= \text{Year-end value} \times \text{Management fee \%} = \$116 \text{ million} \\ &\quad \times 2\% = \$2.32 \text{ million} \\ \text{Incentive fee} &= (\text{Year-end value} - \text{High water mark}) \times \text{Incentive fee \%} \\ &= (\$116 \text{ million} - \$106.96 \text{ million}) \times 20\% = \\ &\quad \$1.81 \text{ million.} \\ \text{Total fees} &= \text{Management fee} + \text{Incentive fee} = \$2.32 \text{ million} + \\ &\quad \$1.81 \text{ million} = \$4.13 \text{ million} \\ \text{Net-of-fees return} &= (\text{Year-end value} - \text{Total fees} - \text{Beginning capital position}) / \text{Beginning capital position} \\ &= (\$116 \text{ million} - \$4.13 \text{ million} - \$98 \text{ million}) / \$98 \text{ million} = \mathbf{14.15\%}.\end{aligned}$$

C is incorrect because 11.87% results from calculating the return based on a Year 2 ending capital position (i.e., Year 3 beginning capital position) of \$100 million. Instead, the return should be calculated based on the Year 2 ending capital position of \$98 million, which is net of fees (\$100 million – \$2 million).

$$\begin{aligned}\text{Management fee} &= \text{Year-end value} \times \text{Management fee \%} = \$116 \text{ million} \\ &\quad \times 2\% = \$2.32 \text{ million} \\ \text{Incentive fee} &= (\text{Year-end value} - \text{High water mark}) \times \text{Incentive fee \%} \\ &= (\$116 \text{ million} - \$106.96 \text{ million}) \times 20\% = \\ &\quad \$1.81 \text{ million.} \\ \text{Total fees} &= \text{Management fee} + \text{Incentive fee} = \$2.32 \text{ million} + \\ &\quad \$1.81 \text{ million} = \$4.13 \text{ million} \\ \text{Net-of-fees return} &= (\text{Year-end value} - \text{Total fees} - \text{Beginning capital position}) / \text{Beginning capital position} \\ &= (\$116 \text{ million} - \$4.13 \text{ million} - \$100 \text{ million}) / \$100 \text{ million} = 11.87\%.\end{aligned}$$

B is incorrect because 12.33% results from calculating the incentive fee using the net capital position at the end of Year 2 instead of the high-water mark value established at the end of Year 1.

$$\text{Management fee} = \text{Year-end value} \times \text{Management fee \%} = \$116 \text{ million} \times 2\% = \$2.32 \text{ million}$$

$$\begin{aligned} \text{Incentive fee} &= (\text{Year-end value} - \text{Year 2 Ending capital position}) \times \text{Incentive fee \%} \\ &= (\$116 \text{ million} - \$98 \text{ million}) \times 20\% = \$3.60 \text{ million.} \end{aligned}$$

$$\text{Total fees} = \text{Management fee} + \text{Incentive fee} = \$2.32 \text{ million} + \$3.60 \text{ million} = \$5.92 \text{ million}$$

$$\begin{aligned} \text{Net-of-fees return} &= (\text{Year-end value} - \text{Total fees} - \text{Beginning capital position}) / \text{Beginning capital position} \\ &= (\$116 \text{ million} - \$5.92 \text{ million} - \$98 \text{ million}) / \$98 \text{ million} = \mathbf{12.33\%}. \end{aligned}$$

Introduction to Alternative Investments

113 When a security that was in an upward trend falls 1% below its trendline, a technical analyst will *most likely* determine that:

- A** a downward trend is beginning.
- B** the upward trend is ending.
- C** the trendline needs an adjustment.

C is correct. When a security's price drops through the trendline by a significant amount (at least 5% to 10%), this decline signals that the uptrend has ended and further decline may follow. Minor breakthroughs simply call for the trendline to be moderately adjusted over time. A security falling 1% below its trendline is considered a minor breakthrough.

A is incorrect. The amount the price falls below the trendline would need to be at least 5% before it would be considered a signal that the uptrend is ending and may signal a further decline in price.

B is incorrect. The amount the price falls below the trendline would need to be at least 5% before it would be considered a signal that the uptrend is ending.

Technical Analysis

114 Following its decision to divest its non-core assets, analysts expect HCL Corp's standard deviation of returns to rise to 30% and its correlation with the market portfolio to remain unchanged at 0.8. The risk-free rate and the market risk premium are expected to remain unchanged at 6% and 8%, respectively. The market portfolio's standard deviation of returns, however, is expected to decrease to 15%. The firm's expected return after the restructure is *closest* to:

- A** 9.2%.
- B** 17.6%.
- C** 18.8%.

C is correct. We first compute the firm's beta using

$$\beta_i = \frac{\rho_{i,m}\sigma_i}{\sigma_m}$$

The beta is

$$\beta_i = \frac{0.8(0.30)}{0.15} = 1.6$$

The expected return is computed using

$$E(R_i) = R_f + [E(R_m) - R_f]\beta_i$$

$$\text{So, } E(R_i) = 0.06 + (0.08)1.6 = 18.8\%.$$

A is incorrect because the beta is incorrectly calculated as $[(0.8 \times 0.15)/0.30] = 0.4$, resulting in an expected return of $0.06 + (0.08)0.4 = 9.2\%$.

B is incorrect because the expected return is incorrectly calculated as $0.08 + (0.06)1.6 = 17.6\%$.

Portfolio Risk and Return: Part II

115 Security analysis is *most likely* a part of which step in the portfolio management process?

- A** The feedback step
- B** The execution step
- C** The planning step

B is correct. The execution step of the portfolio management process has three parts: asset allocation, security analysis, and portfolio construction.

A is incorrect because under the planning step, there are two parts: understating the client's needs and preparation of an investment policy statement.

C is incorrect because under the feedback step, there are two parts: portfolio monitoring and rebalancing and performance measurement and reporting.

Portfolio Management, An Overview

116 Which of the following is *most likely* associated with an investor's ability to take risk rather than the investor's willingness to take risk?

- A** The investor has a long investment time horizon.
- B** The investor believes earning excess returns on stocks is a matter of luck.
- C** Safety of principal is very important to the investor.

A is correct. Investment time horizon is an objective factor that measures the investor's ability to take risk.

B is incorrect because luck is a subjective factor that measures willingness to take risk.

C is incorrect because safety of principal is a subjective factor that measures willingness to take risk.

Basics of Portfolio Planning and Construction

117 An investment has a 50% probability of returning 12% and a 50% probability of returning 6%. An investor prefers this uncertain investment over a guaranteed return of 10%. This preference *most likely* indicates that the investor is risk:

- A** seeking.
- B** averse.
- C** neutral.



A is correct. The expected value of the uncertain investment is 9%, which is less than the guaranteed return of 10%. Only a risk-seeking person would be willing to accept this investment.

B is incorrect. A risk-averse person would prefer the guaranteed outcome of 10%.

C is incorrect. A risk-neutral person would prefer the guaranteed outcome of 10%.

Portfolio Risk and Return: Part I

118 John Smith is given two investment options. Option A is a payment with a 50% chance of getting \$100 and a 50% chance of getting \$0. Option B is a guaranteed payment of \$50. If Smith chooses Option A over Option B, his risk preference is *best* described as risk:

- A** seeking.
- B** averse.
- C** neutral.



A is correct. Both options have the same expected return of \$50. Option A, however, has a higher risk (standard deviation) than Option B. Therefore, John Smith's risk preference is that of risk seeking.

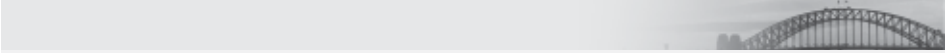
B is incorrect because a risk-averse person would prefer a lower risk profile given the same expected return.

C is incorrect because a risk-neutral person would be indifferent about the risk profile of the outcome.

Portfolio Risk and Return: Part I

119 Risk management is a process that can *most likely* be best described as:

- A** minimizing risks while attempting to maximize returns.
- B** forecasting the level of risk that can meet a defined required return.
- C** defining a level of risk to be taken with the goal of maximizing the portfolio's value.



C is correct. Risk management is the process by which an organization or individual defines the level of risk to be taken, measures the level of risk being taken, and adjusts the latter toward the former, with the goal of maximizing the company's or portfolio's value or the individual's overall satisfaction or utility.

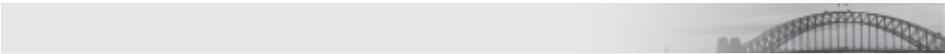
A is incorrect. Risk management is actively understanding how to best balance the achievement of goals with an acceptable chance of failure.

B is incorrect. Risk management is not about predicting risks.

Risk Management: An Introduction

120 "Fintech" is *best* described as:

- A** technology-driven innovation in the financial service industry.
- B** the collection of large quantities of financial data from a variety of sources in multiple formats.
- C** the use of technical models to describe patterns in financial markets and make trading decisions.



A is correct. In its broadest sense, the term "fintech" generally refers to technology-driven innovation occurring in the financial service industry.

B is incorrect. The collection of large quantities of data from a variety of sources in multiple formats is the description of big data.

C is incorrect. The use of technical models to describe patterns in financial markets and make trading decisions is the description of technical analysis.

Fintech in Investment Management
